Scoped O.1 on the Cone

Executed Apex Records, a Closed Constant-Profile Slavnov–Taylor Row, Weighted Apex Coefficients, and Ward-Identity Boundaries

An artifact-verified revision of the v5 proof-level audit

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Abstract

Version 6 is an artifact-verified revision of the v5 audit of the constant-transverse-profile calculation for the scoped O.1 problem on $C_\alpha \times \mathbb{R}^2$. It keeps v5's structure, its evidence classes, and its conditional theorems. What it changes is that the executable record behind v4 is now archived, rerun, and audited, and it adds the following.

New in v6.

(1) *Catalogue re-derived.* Every entry of the v5 coefficient catalogue agrees exactly with the executed spectral reductions of the frozen XVI-P vertex arrays. The executed record also contains three things v4 never printed: the (12)(12) entries $\widehat{B}_{\mathrm{D01}} = \widehat{B}_{\mathrm{D03}} = -8c_1P^2\,\kappa^2/(16\pi)$; the complete D02 tadpole array $\frac{\kappa^2}{2\pi}(c_0 + 2c_1 + c_2)$ times a fixed tensor; and $\widehat{B}_{\mathrm{D12}} = 0$. Mixed-helicity monomials occur only on vector-type keys, which are excluded from the record, not set to zero. The D13 transverse source trace is computed to be zero.

(2) *The hh Slavnov–Taylor failure is diagnosed.* The v4 failure comes from the coordinate weight of the antifield $K'_{(03)}$. The correct weight is -2; v4 scanned only positive weights. With conventions derived from the frozen vertex, the constant-profile hh/Kc row closes on CR and on F. The closure is exact as a rational identity in α, with $\lambda = -2\mathrm{i}$. This value is fixed in advance by the tree-level data, not fitted. UNIT closes trivially.

(3) *A second apex coefficient.* We prove the r^2-weighted apex coefficient

$$\mathcal{D}_b = -\frac{11\alpha}{180} + \frac{B_2(b)}{3\alpha} - \frac{B_4(b)}{6\alpha^3}, \qquad \mathrm{Tr}_{\mathrm{ap}}\big[r^2 f(A_b)\big] = -\mathcal{D}_b f'(0).$$

Its singular-channel shift is $2\nu(1-\nu^2)/3$. It passes exact orbifold and twisted-orbifold checks, and it has no flat CR remnant.

(4) *An exact obstruction for* (H_I) *in D01, and its resolution.* The $S = 2 \to 3$ composite trace has the non-integrable diagonal $r^{-2\nu-2}$, so (H_I) fails as an L^2 statement. Its unique homogeneous (Mellin) continuation, however, vanishes identically in ν. This is shown in closed form and confirmed independently by quadrature.

(5) *Invariant structure of the hh poles.* The constant-profile hh poles decompose into six $SO(1,1)$-invariant structures. In the complete hh sum the purely longitudinal structures cancel, and the gauge variance enters only through $b_6 + b_7 = -\kappa^2 V/(4\pi)$. The CR–F difference is gauge invariant: $-\nu\kappa^2 P^2 f_0 f_0'/(4\pi)$.

A point-by-point table answers every issue v5 raised against v4. An appendix lists full SHA-256 digests and rerun commands.

Still open. The 26 Ward combinations and the 34-dimensional projection; the six exact values; the D04 quadratic-profile part of the second $\phi\phi$ value; the evanescent coefficients; the 18-dimensional engine map; an exact (Span) certificate; and the vector-type keys. None of these

is reported as zero. The scope remains one loop at $O(\kappa^2)$, the massless two-point sector, frozen de Donder gauge, and the XVI-P prescription. The cone and the CR branch remain inputs, and the destructive-readout analogy supplies no premise. `bulk_symmetry_restored` = `false`, and O.1 on the cone remains `NOT_CLOSED`.

Contents

1 Question, scope, and evidence classes **3**
- 1.1 The scoped question . . . 3
- 1.2 What is proved, retained, or conditional . . . 4
- 1.3 What v6 adds . . . 4
- 1.4 Programme non-transfer . . . 5

2 Prescribed cone, fields, and branch domains **5**
- 2.1 Geometry and signatures . . . 5
- 2.2 Scalar and helicity realizations . . . 5
- 2.3 CR selection is not a physical core limit . . . 6
- 2.4 Constant transverse profile is a jet restriction . . . 6

3 A normalized apex heat coefficient **7**
- 3.1 Why an ordinary trace is unavailable . . . 7
- 3.2 Apex coefficients are not only diffracted-wave coefficients . . . 8

4 Singular Bessel channels and the CR heat shift **9**

5 An r^2-weighted apex coefficient (new in v6) **10**
- 5.1 Application: the second $\phi\phi$ combination (partial) . . . 12

6 Conditional spectral reduction of a two-point graph **13**
- 6.1 Helicity classes and one closed transverse trace . . . 13
- 6.2 The missing hypotheses . . . 13
- 6.3 Intertwining is a domain statement . . . 14
- 6.4 The D01 obstruction made exact (new in v6) . . . 15
- 6.5 Longitudinal UV normalization . . . 16

7 The retained constant-profile pole catalogue **16**
- 7.1 Status and notation . . . 16
- 7.2 Scalar two-point sector . . . 17
- 7.3 Tensor two-point sector . . . 17
- 7.4 Tadpoles D02 and D12 and the (12) entries (new in v6) . . . 18
- 7.5 Ghost and source-ghost sector . . . 19

8 Exact checks available from the printed catalogue **19**
- 8.1 A restricted branch-transversality check . . . 20

9 The ghost-equation row and its exact scope **21**
- 9.1 An explicit longitudinal contraction . . . 21
- 9.2 The Maxwell–Friedrichs comparison . . . 22

10 The hh Slavnov–Taylor row: v4 diagnostic, diagnosis, and constant-profile closure **22**
10.1 The supplied attempted row (v4, historical) . . . 22
10.2 Why UNIT requires its own analytic control theorem . . . 22
10.3 Diagnosis of the v4 failure (new in v6) . . . 23
10.4 An *a priori* normalization . . . 23
10.5 Constant-profile closure on CR and F . . . 24

11 Invariant decomposition of the constant-profile hh poles (new in v6) **25**

12 What constant-profile data cannot determine **26**
12.1 The full basis projection . . . 26
12.2 Joint-apex condition (J) . . . 26
12.3 Regularity at special cone openings . . . 26
12.4 Evanescent data and finite linear certificates . . . 27

13 Companion comparisons and finite-report boundaries **27**
13.1 Smooth cores, one-point data, and infrared control . . . 27
13.2 Continuum kernels and echo reports . . . 28
13.3 Finite Enter instruments and their ledgers . . . 28

14 Historical corrections and reproducibility obligations **29**
14.1 Corrections retained from v4 . . . 29
14.2 Proof-level corrections made in v5 (retained) . . . 29
14.3 Corrections in v6 . . . 29
14.4 Artifact inventory (v6) . . . 29
14.5 Status of the computational-certificate obligations . . . 30

15 Point-by-point response to the v5 audit (new in v6) **30**

16 Ledger and release **32**

A Self-adjoint apex dynamics and the readout analogy **34**
A.1 Boundary form and flux . . . 34
A.2 What a loss channel would additionally require . . . 34
A.3 The precise I-C comparison . . . 35
A.4 Why the analogy supplies no O.1 premise . . . 36

B Source and status declarations **36**

C Artifact manifest and rerun commands (new in v6) **37**

References **40**

1 Question, scope, and evidence classes

1.1 The scoped question

The supplied v4 sheet studies transport of the separately scoped XVI-Q record Q11 to a prescribed conical product, with:

(i) one-loop order and coupling order $O(\kappa^2)$;

(ii) a two-point sector and vanishing declared scalar mass;

(iii) frozen de Donder gauge;

(iv) dimensional reduction with quasi-d loop momenta and quasi-four-dimensional vertex indices;

(v) the frozen XVI-P prescription and the p22 contract;

(vi) curvature-regular (CR) and componentwise Friedrichs (F) apex realizations compared without selecting either by readout.

XVI-P, XVI-Q, and XVI-R are frozen comparison records [3, 4, 5]. The later supplied p23 excerpt is not silently substituted for the p22 contract.

The principal issue is not whether a conical heat coefficient exists. It is whether the complete differentiated graph, on its declared domain and in its declared prescription, reduces to that coefficient and obeys the required interacting identities.

1.2 What is proved, retained, or conditional

This revision uses four evidence classes. v6 adds the last one.

Locally proved.
The normalized scalar apex coefficient; the singular-channel heat shift; the new r^2-weighted coefficient and its shift; a conditional graph-reduction theorem; the longitudinal UV normalization; algebraic identities between the printed polynomials; and the stated access/flux distinctions.

Executed and artifact-verified in v6.
Statements carrying the tag [exec]. Each was produced by a program in `code/` and logged in `logs/v4/` or `logs/v6/`, and was rerun or checked in this revision. Its full SHA-256 digest is listed in Appendix C. The vertex inputs are read-only copies of the frozen XVI-P/Q files; their digests equal the originals. Exact statements are exact rational or symbolic identities in α or ν. Numerical statements give their precision. An executed result is a reproducible computation, not an independent review.

Supplied computational record.
v4 reports that v6 did not rerun, together with the historical registry intake. These are kept as reports.

Conditional graph identification.
Equality of a printed polynomial with a particular frozen graph integral still needs the graph-specific hypotheses of Assumption 6.1. v6 discharges several of these for the constant-profile record; the ledger in Section 16 lists them. No proof is reconstructed from a routine name.

Submission and proof boundary (v6)

v6 now archives the vertex arrays, routing, prefactors, and logs that v5 found missing. The constant-profile catalogue is therefore an executed result on the archived frozen inputs, and no longer just a supplied report. It is still not an independently refereed computation. The spectral reduction of each graph remains conditional on Assumption 6.1. Only (R1) and parts of (R3) are discharged here (Sections 7, 6.4). (R2), (R4), and (R5) are still the stated hypotheses.

The coefficient values printed in v4 and v5 are unchanged. v6 adds entries but alters none.

1.3 What v6 adds

(a) *hh Slavnov–Taylor row* (Section 10). Derived conventions, an *a priori* constant λ, and exact closure on CR and F for the constant profile. The v4 failure is diagnosed, and UNIT closes.

(b) *Higher apex coefficient* (Section 5). The r^2-weighted coefficient $\mathcal{D}_b$, with checks. The quadratic-profile $\phi\phi$ value is partial: the D05a part is 0 and the D04 part is open.

(c) ($\mathrm{H}_{I,\mathrm{D01}}$) (Section 6.4). The exact L^2 obstruction, and an exact vanishing Mellin value.

(d) *Evanescent coefficients* (Section 12). A pole-level statement only; the coefficients themselves are open.

(e) *hh map* (Section 11). Six invariant structures, the complete D02/D12 arrays, and the coefficient vectors. The 18-dimensional engine map is open.

(f) (Span) *certificate* (Section 12). Open, with the precise reason.

(g) *Response table* (Section 15). Every v5 issue, answered point by point.

1.4 Programme non-transfer

Connections to I-B, I-B-G, I-B-G-S, I-B-G-S-B, I-B-VP, and I-C are separations, not additional analytical premises or a declared path. They do not discharge O.1, open the downstream XV-B admission exit, or resolve the unrelated H2/I3/J2 characteristic-product obligations.

The historical H2 ending in XIV-H is retained version-specifically; the narrower I2 internal-amplitude construction in XIV-I is not withdrawn. The complete joint maps at I3/J2 and their subsequent admission chain remain separate [20, 21]. Missing values are never set to zero.

2 Prescribed cone, fields, and branch domains

2.1 Geometry and signatures

For the Euclidean spectral analysis use

$$X_\alpha = C_\alpha \times \mathbb{R}^2, \qquad \mathrm{d}s_E^2 = \mathrm{d}r^2 + \alpha^2 r^2 \mathrm{d}\varphi^2 + \mathrm{d}y_0^2 + \mathrm{d}y_3^2, \quad 0 \le \varphi < 2\pi, \quad 0 < \alpha \le 1. \tag{2.1}$$

The singular stratum is the codimension-two set

$$\mathcal{W} = \{r = 0\} \times \mathbb{R}^2.$$

Its transverse measure is

$$\mathrm{d}V_\perp = \alpha r \, \mathrm{d}r \, \mathrm{d}\varphi.$$

For the Lorentzian polynomial catalogue the tangent signature is $\eta = \mathrm{diag}(1, -1, -1, -1)$, and

$$P^2 = q_0^2 - q_3^2. \tag{2.2}$$

The symbols q_0, q_3 first denote the two polynomial variables printed in v4. Their conversion to covariant or contravariant momentum components must be fixed when a Ward contraction is made; Section 9 does so explicitly.

The apex is a prescribed conical worldsheet, not an event horizon. No state-dependent metric update is present [6, 8].

2.2 Scalar and helicity realizations

A scalar channel with fractional twist $b \in [0, 1)$ has radial Bessel orders

$$\nu_n = \frac{|n + b|}{\alpha}, \qquad n \in \mathbb{Z}.$$

The Friedrichs realization chooses the regular branch r^{ν_n}, with the no-logarithm branch at $\nu_n = 0$.

For a geometric helicity S, the fractional label is

$$b_{\mathrm{F}}(S) = \{S\alpha\}, \tag{2.3}$$

with fractional part in $[0, 1)$. In the interval $1/2 < \alpha < 1$, the distinct labels for $S = 0, \pm 1, \pm 2$ are

$$0, \quad \alpha, \quad 1-\alpha, \quad 2\alpha - 1, \quad 2 - 2\alpha.$$

For the CR comparison, the v4 labels are

$$b_{\mathrm{CR}}(S) = \begin{cases} S\alpha, & S \geq 0, \\ 1 + S\alpha, & S < 0. \end{cases} \tag{2.4}$$

For $S = \pm 2$ and $1/2 < \alpha < 1$, these labels lie outside the fractional interval. They are not ordinary Aharonov–Bohm holonomies modulo one. They encode an additional endpoint choice.

Put

$$\nu = 2 - \alpha^{-1} \in (0, 1). \tag{2.5}$$

In each exceptional weight-2 channel, CR replaces the regular Bessel realization of order ν by the scale-invariant singular realization with generalized eigenfunction $J_{-\nu}$. Both are nonnegative self-adjoint Bessel realizations in this range [26]. Their difference, not a periodic relabeling alone, produces the heat shift proved in Section 4.

Remark 2.1 (Parameter range of the printed polynomials). The common rational forms used below describe the sector $1/2 < \alpha < 1$, with their continuous values at $\alpha = 1/2$. The F expressions also have their ordinary flat value at $\alpha = 1$.

For $0 < \alpha < 1/2$, the appropriate Friedrichs weight-2 label is 2α, not $2\alpha - 1$. Thus the displayed F polynomials must not be advertised as one unchanged formula on every $0 < \alpha \leq 1$. The scalar heat theorem itself holds for every positive α.

2.3 CR selection is not a physical core limit

The supplied I-B-G-S excerpts describe CR as a mathematical domain selection restoring the stated source-test de Donder compatibility and packet curvature regularity in the exceptional sector [9, `def:geo`, `thm:selection`, `thm:spin2`]. That selection is not a physical smooth-core theorem.

The supplied I-B-G-S-B description reports that its declared nonnegative tensor connection-Laplacian and Lichnerowicz-form families, with the metric spin connection, select a different positive-core limit. In its neutral exceptional spin-2 channels for $\alpha > 1/2$, the description excludes the idealized CR branch in the Mosco/strong-resolvent setting and retains the corresponding source-test defect [10, `cor:exclude`, `thm:contrast`, `prop:warddefect`]. This is a source-qualified comparison; a complete independent tensor proof is not asserted here.

Consequently, CR remains an input to this computation. Analytical angular-tail, complex-kernel, and Cauchy estimates are also distinct from an executed, certified full-kernel run [10, `lem:modetail`, `lem:complexbound`, `thm:Cauchy`].

Separation boundary. The cone, extension, regulator, and external-field class are prescribed. A free inverse, operator limit, one-point finite part, finite instrument, or documentary comparison is not an interacting two-point Ward theorem. No Einstein dynamics, physical tensor Hessian, spacetime emergence, or inherited quantum admission follows.

2.4 Constant transverse profile is a jet restriction

A constant-profile external field is understood as a local jet: it is constant in the transverse variables near the apex, with longitudinal Fourier dependence. It need not be a globally square-integrable constant section of an infinite cone.

Only helicity-zero constant profiles are invariant under the transverse identification. For a symmetric tensor, the permitted transverse scalar is the trace combination

$$h_{11} = h_{22} = f_0.$$

The individual coordinates h_{11} and h_{22} are not separately helicity-zero eigencomponents.

For comparison with v4, the component keys are

$$0 \leftrightarrow 00, \quad 3 \leftrightarrow 03, \quad 9 \leftrightarrow 33, \quad 4 \leftrightarrow 11, \quad 7 \leftrightarrow 22.$$

The supplied component array may retain entries in these coordinates; a genuine constant-profile scalar test uses the transverse trace combination. In particular, if $B_{44} = B_{77}$,

$$B_{f_0 f_0} = B_{44} + 2B_{47} + B_{77} = 2(B_{44} + B_{47}). \tag{2.6}$$

No omitted non-lattice component is inferred to vanish.

3 A normalized apex heat coefficient

3.1 Why an ordinary trace is unavailable

The heat operator on the infinite cone has infinite ordinary trace. Its bulk volume term must be subtracted before an apex coefficient is stated. We use an explicit radial Abel subtraction.

Let A_b^{F} be the nonnegative twisted Friedrichs transverse Laplacian. Its diagonal heat kernel is

$$K_b^{\mathrm{F}}(s; r, \varphi; r, \varphi) = \frac{e^{-z}}{4\pi\alpha s} \sum_{n \in \mathbb{Z}} I_{|n+b|/\alpha}(z), \qquad z = \frac{r^2}{2s}, \quad s > 0. \tag{3.1}$$

This is the Bessel heat representation, with normalization relative to $\mathrm{d}V_\perp$. The required Bessel integral formulas are standard [25, 26]; the subtraction and constant are computed below.

Definition 3.1 (Abel apex trace)**.** For $u > 0$, define

$$\begin{aligned} T_b(u) &= \int_{C_\alpha} e^{-(\cosh u - 1) r^2/(2s)} K_b^{\mathrm{F}}(s; x, x)\, \mathrm{d}V_\perp, \\ T_{\mathrm{bulk}}(u) &= \int_{C_\alpha} e^{-(\cosh u - 1) r^2/(2s)} \frac{\mathrm{d}V_\perp}{4\pi s} = \frac{\alpha}{2(\cosh u - 1)}. \end{aligned} \tag{3.2}$$

When the limit exists, put

$$\operatorname{Tr}_{\mathrm{ap}} e^{-sA_b^{\mathrm{F}}} = \lim_{u \downarrow 0} \bigl(T_b(u) - T_{\mathrm{bulk}}(u)\bigr).$$

This definition specifies a bulk-subtracted scale-invariant coefficient, not an ordinary trace on $L^2(C_\alpha)$.

Theorem 3.2 (Twisted Friedrichs apex constant)**.** *For every $\alpha > 0$, $0 \le b < 1$, and $s > 0$,*

$$\operatorname{Tr}_{\mathrm{ap}} e^{-sA_b^{\mathrm{F}}} = \mathcal{C}_b^{\mathrm{F}} := \frac{1}{12}\left(\frac{1}{\alpha} - \alpha\right) - \frac{b(1-b)}{2\alpha}. \tag{3.3}$$

The result is independent of the proper time s. For

$$f(\lambda) = \int_a^b w(s) e^{-s\lambda}\, \mathrm{d}s, \qquad 0 < a < b < \infty, \quad w \in L^1([a, b]),$$

the corresponding heat-superposition apex functional satisfies

$$\operatorname{Tr}_{\mathrm{ap}} f(A_b^{\mathrm{F}}) = \mathcal{C}_b^{\mathrm{F}} f(0). \tag{3.4}$$

The subscript b on the operator denotes the twist; the integration endpoints in this statement are merely positive proper-time bounds.

Proof. Integrating (3.1) in the angular and radial variables gives

$$T_b(u) = \frac{1}{2} \sum_{n \in \mathbb{Z}} \int_0^\infty e^{-z \cosh u} I_{|n+b|/\alpha}(z) \, \mathrm{d}z.$$

For $u > 0$ and $\rho > -1$,

$$\int_0^\infty e^{-z \cosh u} I_\rho(z) \, \mathrm{d}z = \frac{e^{-\rho u}}{\sinh u}.$$

The nonnegative summands justify the interchange. The geometric series gives

$$\sum_{n \in \mathbb{Z}} e^{-|n+b|u/\alpha} = \frac{\cosh((1/2 - b)u/\alpha)}{\sinh(u/(2\alpha))}.$$

Hence

$$T_b(u) = \frac{\cosh((1/2 - b)u/\alpha)}{2 \sinh u \, \sinh(u/(2\alpha))}. \tag{3.5}$$

With $B_2(b) = b^2 - b + 1/6$, expansion at zero yields

$$T_b(u) = \frac{\alpha}{u^2} + \frac{B_2(b)}{2\alpha} - \frac{\alpha}{6} + O(u^2),$$

whereas

$$T_{\mathrm{bulk}}(u) = \frac{\alpha}{u^2} - \frac{\alpha}{12} + O(u^2).$$

Subtracting proves

$$\mathcal{C}_b^{\mathrm{F}} = \frac{B_2(b)}{2\alpha} - \frac{\alpha}{12},$$

which is (3.3).

The radial change of variable removed s, proving exact proper-time independence. For a compact proper-time interval, the closed expressions and their subtracted limits are uniformly bounded. Dominated convergence and linearity then give

$$\mathrm{Tr}_{\mathrm{ap}} \, f(A_b^{\mathrm{F}}) = \mathcal{C}_b^{\mathrm{F}} \int_a^b w(s) \, \mathrm{d}s = \mathcal{C}_b^{\mathrm{F}} f(0).$$

□

The untwisted coefficient and conical heat-trace methodology are classical [22, 23, 24]. The proof fixes the normalization rather than attributing a numerical re-derivation to unavailable software.

Remark 3.3 (Function class). Theorem 3.2 does not assert an unrestricted ordinary-trace identity for every Schwartz function. Extending (3.4) to resolvent products, differentiated kernels, or meromorphic graph families requires the corresponding functional-calculus and interchange hypotheses. Those hypotheses are stated in Section 6.

3.2 Apex coefficients are not only diffracted-wave coefficients

The scalar Sommerfeld–Carslaw representation, extension-dependent Green identities, and supported transfer in I-B-G concern continuum operators on their stated domains [8, `thm:green`, `eq:Carslaw`, `thm:transfer`]. They do not identify a loop residue with a finite detector output.

One further distinction is necessary: the apex heat constant can receive contributions from geometric images even when a chosen diffracted wave term vanishes.

Example 3.4 (A nonzero apex constant in a pure-image case)**.** For the neutral cone $\alpha = 1/2$,

$$\mathcal{C}_0^{\mathrm{F}} = \frac{1}{8}.$$

The cone is a two-fold rotation quotient. The nontrivial image has plane heat kernel

$$\frac{1}{4\pi s}\exp(-r^2/s)$$

on the diagonal after a half-turn. Integrating over the wedge of angle π gives

$$\frac{\pi}{4\pi s}\int_0^\infty r e^{-r^2/s}\,\mathrm{d}r = \frac{1}{8}.$$

Thus this apex coefficient is present without needing a separately nonzero diffracted-front term.

The finite-window echo ratios of I-B-ECHO and I-C concern a different functional. Their additive front expansion and their zero-deficit witness on $1/2 < \alpha \le 1$ do not turn $\mathcal{C}_b$ into that ratio [11, 19].

Separation boundary. The cone, extension, regulator, and external-field class are prescribed. A free inverse, operator limit, one-point finite part, finite instrument, or documentary comparison is not an interacting two-point Ward theorem. No Einstein dynamics, physical tensor Hessian, spacetime emergence, or inherited quantum admission follows.

4 Singular Bessel channels and the CR heat shift

For $0 < \nu < 1$, let A_ν^+ and A_ν^- denote the scale-invariant regular and singular nonnegative self-adjoint radial realizations in $L^2((0,\infty), r\,\mathrm{d}r)$. Their radial heat kernels are

$$k_\nu^\pm(s; r, r') = \frac{1}{2s}e^{-(r^2+r'^2)/(4s)} I_{\pm\nu}\left(\frac{rr'}{2s}\right).$$

These choices are endpoint realizations, not ordinary potentials added to the regular cone.

Theorem 4.1 (One-channel flip)**.** *For $0 < \nu < 1$,*

$$\mathrm{Tr}_{\mathrm{rad}}\big(e^{-sA_\nu^-} - e^{-sA_\nu^+}\big) = \nu, \qquad s > 0. \tag{4.1}$$

The diagonal relative integral is convergent. Consequently, replacing one such regular angular channel by its singular realization increases its apex heat constant by ν.

Proof. The radial trace difference is

$$\frac{1}{2}\int_0^\infty e^{-z}\big(I_{-\nu}(z) - I_\nu(z)\big)\,\mathrm{d}z.$$

Near zero, the worst term is $z^{-\nu}$, which is integrable because $\nu < 1$. At infinity use

$$I_{-\nu}(z) - I_\nu(z) = \frac{2\sin(\pi\nu)}{\pi}K_\nu(z).$$

The resulting integrand decays exponentially [25].

Insert the Abel factor and use the Laplace transform from the preceding proof. The integral becomes

$$\frac{e^{\nu u} - e^{-\nu u}}{2\sinh u} = \frac{\sinh(\nu u)}{\sinh u}.$$

Dominated convergence gives the limit ν. Scale invariance makes it independent of s. □

Corollary 4.2 (Three apex coefficients in the comparison sector)**.** *For* $1/2 < \alpha < 1$*, define*

$$\begin{aligned} c_0 &= \frac{1-\alpha^2}{12\alpha}, \\ c_1 &= \frac{5\alpha^2-6\alpha+1}{12\alpha}, \\ c_2^{\mathrm{F}} &= \frac{23\alpha^2-36\alpha+13}{12\alpha}, \\ c_2^{\mathrm{CR}} &= \frac{23\alpha^2-12\alpha+1}{12\alpha}. \end{aligned} \tag{4.2}$$

These are the per-helicity apex coefficients for weight zero, weight one, and the two respective weight-two realizations. They satisfy

$$c_1 - c_0 = \frac{\alpha-1}{2}, \tag{4.3}$$

$$c_2^{\mathrm{CR}} - c_0 = 2\alpha - 1, \tag{4.4}$$

$$c_2^{\mathrm{F}} - c_0 = \frac{(\alpha-1)(2\alpha-1)}{\alpha}, \tag{4.5}$$

$$c_2^{\mathrm{CR}} - c_2^{\mathrm{F}} = 2 - \frac{1}{\alpha} = \nu. \tag{4.6}$$

At $\alpha = 1/2$*, continuous evaluation gives*

$$c_0 = c_2^{\mathrm{F}} = c_2^{\mathrm{CR}} = \frac{1}{8}, \qquad c_1 = -\frac{1}{8}.$$

Proof. Apply (3.3) at $b = 0, \alpha, 2\alpha - 1$. For the exceptional CR channel add ν from Theorem 4.1. Expansion gives (4.2); subtraction gives the remaining identities. The mirror labels have the same coefficients because $b(1-b) = (1-b)b$. □

The polynomial expression obtained by inserting $b = 2\alpha$ into (3.3) reproduces c_2^{CR}. This is now justified by a specified channel flip. It is not a general license to use arbitrary real b as a fractional twist.

Remark 4.3 (The CR flat-limit remnant). As $\alpha \uparrow 1$,

$$c_0 \to 0, \qquad c_1 \to 0, \qquad c_2^{\mathrm{F}} \to 0, \qquad c_2^{\mathrm{CR}} \to 1.$$

The last statement is a one-sided limit of the singular-channel family. At $\alpha = 1$, its formal order is $\nu = 1$, and

$$\int_0^{\epsilon} r\,|r^{-1}|^2\,\mathrm{d}r = \infty.$$

The deficient singular L^2 branch used above no longer exists in the same form. Therefore a nonzero limiting CR polynomial is not, without an additional theorem, the heat trace of a CR realization at the exactly smooth endpoint. Nonuniform spin-dependent flat limits are also discussed in the established conical heat-kernel literature [23].

5 An r^2-weighted apex coefficient (new in v6)

Constant transverse profiles see only $\mathcal{C}_b$. A quadratic transverse jet $G = r^2 g$ needs the next moment of the apex heat distribution. v5 recorded that no value of the higher coefficient $\mathcal{D}_b$ had been supplied or derived. This section derives it with the same Abel subtraction used in Definition 3.1.

Definition 5.1 (Weighted Abel apex trace)**.** For $u > 0$ put

$$T_b^{(2)}(u) = \int_{C_\alpha} r^2 e^{-(\cosh u - 1)r^2/(2s)} K_b^{\mathrm{F}}(s; x, x)\, \mathrm{d}V_\perp,$$
$$T_{\mathrm{bulk}}^{(2)}(u) = \int_{C_\alpha} r^2 e^{-(\cosh u - 1)r^2/(2s)} \frac{\mathrm{d}V_\perp}{4\pi s} = \frac{\alpha s}{(\cosh u - 1)^2}.$$

When the limit exists, set $\mathrm{Tr}_{\mathrm{ap}}[r^2 e^{-sA_b}] = \lim_{u \downarrow 0}(T_b^{(2)}(u) - T_{\mathrm{bulk}}^{(2)}(u))$.

Theorem 5.2 (Weighted twisted Friedrichs apex coefficient)**.** *Let $\alpha > 0$, $0 \le b < 1$, and $s > 0$. With Bernoulli polynomials $B_2(b) = b^2 - b + \frac{1}{6}$ and $B_4(b) = b^4 - 2b^3 + b^2 - \frac{1}{30}$,*

$$\mathrm{Tr}_{\mathrm{ap}}\big[r^2 e^{-sA_b^{\mathrm{F}}}\big] = \mathcal{D}_b\, s, \qquad \mathcal{D}_b = -\frac{11\alpha}{180} + \frac{B_2(b)}{3\alpha} - \frac{B_4(b)}{6\alpha^3}. \tag{5.1}$$

Equivalently,

$$\mathcal{D}_b = \frac{1 + 10\alpha^2 - 11\alpha^4 - 60\alpha^2 b(1-b) - 30b^2(1-b)^2}{180\alpha^3}. \tag{5.2}$$

For a heat superposition f as in Theorem 3.2,

$$\mathrm{Tr}_{\mathrm{ap}}[r^2 f(A_b^{\mathrm{F}})] = -\mathcal{D}_b\, f'(0). \tag{5.3}$$

Proof. With $z = r^2/(2s)$, we have $r^2 = 2sz$ and $r\,\mathrm{d}r = s\,\mathrm{d}z$. The angular integral and the heat representation (3.1) then give

$$T_b^{(2)}(u) = s \sum_{n \in \mathbb{Z}} \int_0^\infty z\, e^{-z\cosh u} I_{|n+b|/\alpha}(z)\, \mathrm{d}z.$$

Differentiate the Laplace transform from the proof of Theorem 3.2 with respect to u:

$$\int_0^\infty z\, e^{-z\cosh u} I_\rho(z)\, \mathrm{d}z = -\frac{1}{\sinh u}\frac{\mathrm{d}}{\mathrm{d}u}\frac{e^{-\rho u}}{\sinh u}, \qquad \rho > -1.$$

Since all summands are nonnegative, the sum may be interchanged with the integral, giving

$$\frac{T_b^{(2)}(u)}{s} = -\frac{1}{\sinh u}\frac{\mathrm{d}}{\mathrm{d}u}\left[\frac{\cosh((\frac{1}{2} - b)u/\alpha)}{\sinh u\, \sinh(u/(2\alpha))}\right].$$

Laurent expansion at $u = 0$ shows that the $u^{-4}, u^{-3}, u^{-2}, u^{-1}$ terms of $T_b^{(2)}/s$ and $T_{\mathrm{bulk}}^{(2)}/s = \alpha/(\cosh u - 1)^2$ coincide. The difference of their constant terms is (5.1). Expanding the Bernoulli polynomials gives (5.2).

For a heat superposition $f(\lambda) = \int w(s) e^{-s\lambda} \mathrm{d}s$, dominated convergence on the compact proper-time support gives $\mathrm{Tr}_{\mathrm{ap}}[r^2 f(A)] = \mathcal{D}_b \int s\, w(s)\, \mathrm{d}s = -\mathcal{D}_b f'(0)$. [exec] The Laurent algebra is carried out in `code/weighted_apex.py`: all four pole terms vanish identically, and the constant term equals (5.1) (`logs/v6/weighted_checks.json`, key `D_matches_log`). □

Proposition 5.3 (Weighted singular-channel shift)**.** *For $0 < \nu < 1$, replacing one regular channel of order ν by its singular realization changes the weighted apex coefficient by*

$$\Delta\mathcal{D} = \frac{2\nu(1 - \nu^2)}{3}.$$

For the CR weight-two channel, $\nu = 2 - \alpha^{-1}$ and

$$\mathcal{D}_{S=2}^{\mathrm{CR}} = \mathcal{D}_{2\alpha-1} + \Delta\mathcal{D} = \mathcal{D}_{b=2\alpha},$$

the polynomial (5.2) evaluated at $b = 2\alpha$.

Proof. The weighted analogue of the proof of Theorem 4.1 replaces $e^{-\nu u}$ by $e^{+\nu u}$ in one channel. The difference is $-(\sinh u)^{-1}\partial_u[2\sinh(\nu u)/\sinh u]$. Its limit as $u \to 0$ is $-u^{-1}\partial_u[2\nu(1+(\nu^2-1)u^2/6)] \to 2\nu(1-\nu^2)/3$. Near zero the singular kernel is $O(z^{1-\nu})$, so the weighted diagonal difference is absolutely integrable. [exec] The identity $\mathcal{D}_{2\alpha} - \mathcal{D}_{2\alpha-1} = \Delta\mathcal{D}$ is checked exactly as a rational function of α (`logs/v6/weighted_checks.json`). □

Corollary 5.4 (Comparison-sector values)**.** *For $1/2 < \alpha < 1$:*

$$\begin{aligned}
\mathcal{D}_0 &= \frac{(1-\alpha^2)(1+11\alpha^2)}{180\alpha^3},\\
\mathcal{D}_{S=1} &= \frac{(\alpha-1)(\alpha+1)(19\alpha^2-1)}{180\alpha^3},\\
\mathcal{D}^{\mathrm{F}}_{S=2} &= -\frac{(\alpha-1)(251\alpha^3-829\alpha^2+601\alpha-119)}{180\alpha^3},\\
\mathcal{D}^{\mathrm{CR}}_{S=2} &= -\frac{(\alpha-1)(251\alpha^3-109\alpha^2+\alpha+1)}{180\alpha^3}.
\end{aligned}$$

All four vanish as $\alpha \uparrow 1$. In particular, the weighted CR coefficient has no *flat remnant, unlike $c_2^{\mathrm{CR}} \to 1$ (Remark 4.3), because $\Delta\mathcal{D} \to 0$ as $\nu \to 1$.*

Proposition 5.5 (Orbifold checks)**.** *For $\alpha = 1/N$ with $N \in \{2,\dots,8\}$,*

$$\mathcal{D}_0(1/N) = \frac{1}{4N}\sum_{k=1}^{N-1}\csc^4\frac{\pi k}{N} = \frac{(N^2-1)(N^2+11)}{180N}.$$

This gives 1/8, 8/27, 9/16, 24/25, 329/216, 16/7, 105/32*. For the twisted orbifolds $(N,j) = (3,1),(4,1),(5,2)$, with $b = j/N$, the image sums weighted by holonomy phases equal $\mathcal{D}_b$, namely* $-4/27$, $-1/16$, *and* $-16/25$.

Proof. On $\mathbb{R}^2/\mathbb{Z}_N$ the heat kernel is the image sum. The k-th image contributes $(4\pi s)^{-1}e^{-r^2\sin^2(\pi k/N)/s}$. Integrating r^2 times it over the wedge of angle $2\pi/N$ gives $s/(4N\sin^4(\pi k/N))$. The identity element is the bulk term. [exec] Both sides are evaluated exactly for each N, and to 10^{-12} for the twisted cases. See the key `orbifold_csc4` in `logs/v6/weighted_checks.json`, and `logs/v6/weighted_apex.json`. □

Remark 5.6 (Literature comparison). The untwisted moment $\mathcal{D}_0$ has the $\csc^4$-sum structure of the classical conical heat coefficients [22]. The identification of $2\pi\alpha\,\mathcal{D}_0$ with the C_4-type coefficient of that reference is made from memory of its normalization (†). It is not used as an input: the orbifold identity above is self-contained.

Two literature results are analogues of the CR one-sided remnant of c_2^{CR}: the non-uniform spin-dependent flat limits of [23], and the spin-one horizon contact term of [32]. Corollary 5.4 shows that this remnant does *not* propagate to the r^2 moment. No numerical spin-2 coefficient is imported from either work.

5.1 Application: the second $\phi\phi$ combination (partial)

By (5.3), a quadratic transverse jet contributes $-\mathcal{D}_S$ times the derivative, at zero transverse spectral parameter, of the longitudinal integrand. Within Assumption 6.1:

- [exec] *D05a.* After the weighted insertion, the single-line tadpole has longitudinal integrand proportional to $p^2\mathcal{D}_S/l^4$. At $D = 2$ this is $O(R^{-4})$ at large radius, so it has no logarithmic UV term (Lemma 6.6). The D05a contribution to the quadratic-profile pole is therefore *computed* to be 0.

- *D04.* The bubble couples a graviton line and a scalar line in different channels. Its r^2 insertion requires mixed-channel traces of the transverse position operators $x_\pm$ between channels S and $S\pm1$. These are not functions of a single A_S, so they fall outside Theorem 5.2 and outside (R2). The D04 part is OPEN.

The second scalar basis value is therefore *partial*. It is not reported as equal to its D05a part.

6 Conditional spectral reduction of a two-point graph

6.1 Helicity classes and one closed transverse trace

Under longitudinal/transverse splitting, a symmetric four-dimensional tensor has:

- four weight-zero components: the symmetric longitudinal block and the transverse trace;
- four weight-one components: the longitudinal–transverse block;
- two weight-two components: the transverse traceless block.

A vector has two weight-zero and two weight-one components. A scalar has weight zero.

These are geometric helicities. The older labels L, M, T do not, without their definitions, determine the correct apex weight.

A reduced closed loop has one transverse spectral trace. Accordingly, a contribution normalized to one common spectral channel receives one coefficient c_S. It does not receive the product of one apex coefficient for every propagator in that trace.

6.2 The missing hypotheses

Assumption 6.1 (Graph-specific spectral reduction contract)**.** For a specified graph G, external constant helicity-zero jet, and fixed routing, require:

(R1) **Complete vertex data.** The tensor vertices, propagator numerators, source insertions, statistics signs, symmetry factors, and independent-component normalizations are specified.

(R2) **Operator reduction.** After the complete vertex contraction, each contributing transverse operator expression is reduced to a finite sum

$$f_{G,S}(A_S; l, p),$$

on one identified channel domain, plus a remainder proved not to contribute to the apex UV residue.

(R3) **Domain and trace legitimacy.** Every differentiated intertwiner, integration by parts, and trace rearrangement is valid on that domain. Any apex boundary term or finite-part trace defect is included, or proved to vanish after the full permitted sum.

(R4) **UV/IR separation.** An auxiliary infrared prescription, such as a positive mass in the Schwinger representation, distinguishes the UV residue from massless infrared poles. A scaleless integral is not assigned its UV pole merely by ignoring its infrared part.

(R5) **Meromorphic interchange.** The heat-superposition apex identity extends to these regulated spectral functions. The transverse subtraction, longitudinal integration, removal of auxiliary cutoffs, and Laurent-residue operation commute in the stated meromorphic prescription.

A graph is called *spectrally certified in this sector* only after these conditions have been established for that graph.

Theorem 6.2 (Conditional spectral apex reduction)**.** *Under Assumption 6.1, the apex UV residue of G is*

$$\operatorname{Res}_{\mathrm{UV}} G_{\mathrm{ap}}(p) = \sum_S c_S \operatorname{Res}_{\mathrm{UV}} \left[\mu^{4-d} \int \frac{\mathrm{d}^{\,d-2} l_\parallel}{(2\pi)^{d-2}} f_{G,S}(0; l_\parallel, p) \right]. \tag{6.1}$$

Multiplicity, statistics, coupling, and vertex phases are included in the corresponding $f_{G,S}$. For a flat-integrand representation this is the $l_\perp = 0$ expression only when the operator reduction (R2)–(R3) *has actually established that identification.*

Proof. First retain the auxiliary proper-time and infrared cutoffs. By (R2), the complete graph is a finite sum of functions of the identified A_S, up to the residue-free remainder. Apply (3.4) and Theorem 4.1 to each regularized heat-superposition term. This replaces its transverse apex trace by $c_S f_{G,S}(0; l, p)$.

Condition (R3) ensures that the transformations used to reach these terms have not omitted a boundary or trace defect. Condition (R5) then allows longitudinal integration, cutoff removal, and residue extraction in that order or the stated equivalent order. The remainder contributes zero by (R2). This gives (6.1). □

Remark 6.3 (What the heat theorem does not prove). A bubble containing two different channel realizations, a domain-changing differential operator, or an unaccounted boundary term is not automatically a function of one A_S. The phrase "the external transverse momentum is zero" does not establish (R2) or (R3). Theorem 6.2 therefore cannot be replaced by the unqualified statement that every differentiated two-point graph equals its flat integrand at $l_\perp = 0$.

6.3 Intertwining is a domain statement

The formal Bessel recurrences give differential identities between solutions. Equality of self-adjoint resolvents additionally requires the differentiated solution to lie in the target domain [25, 26].

For example, if a singular leading term behaves as $r^{-\nu}$, one transverse derivative can produce $r^{-\nu-1}$. For $\nu > 0$,

$$\int_0^\epsilon r\, |r^{-\nu-1}|^2 \, \mathrm{d}r = \int_0^\epsilon r^{-2\nu-1} \, \mathrm{d}r = \infty.$$

Thus a formal recurrence is not an unrestricted L^2 intertwiner through the next helicity.

The supplied v4 record reports:

- D03, D04, and D11 as clean in its constant-profile reduction;
- D06 and D13 as clean on CR;
- D01 as conditional on (H_I), because its derivative tower can reach the $S = 2 \to 3$ sector.

These are retained as graph-assessment reports, not replaced by an unavailable local derivation.

For D01, the unrestricted tower version of (H_I) is not an available hypothesis: the same record states its failure. A nonvacuous successor must prove that the *actual contracted graph* avoids the forbidden terms, includes their boundary contributions, or cancels them in the complete sum. We denote that graph-local requirement by

$$(\mathrm{H}_{I,\mathrm{D01}}).$$

Merely assuming the already excluded unrestricted tower would not certify D01.

6.4 The D01 obstruction made exact (new in v6)

Let $\nu = 2 - \alpha^{-1} \in (0,1)$. In the flipped weight-two channel, the CR–F difference of radial heat kernels is

$$\Delta k(s;r,r') = k_\nu^- - k_\nu^+ = \frac{1}{2s} e^{-(r^2+r'^2)/(4s)} \, \frac{2\sin\pi\nu}{\pi} \, K_\nu\!\left(\frac{rr'}{2s}\right).$$

The D01 derivative tower carries this channel into $S = 3$ through the raising operator

$$X = \partial_r - \frac{\nu}{r},$$

the radial part of the helicity-raising derivative on the $J_{-\nu}$ branch. The term that ($\mathrm{H}_{I,\mathrm{D01}}$) must control is the composite diagonal

$$\mathcal{R}(s;r) = \left[X_r X_{r'} \Delta k(s;r,r')\right]_{r'=r}.$$

Theorem 6.4 (Exact obstruction and Mellin value for D01)**.**

(i) Obstruction. *As $r \to 0$, $\mathcal{R}(s;r) = c_* \, r^{-2\nu-2} + O(r^{-2\nu})$, with $c_* \neq 0$ for every $0 < \nu < 1$. Hence $\int_0^\epsilon r\,\mathcal{R}\,\mathrm{d}r$ diverges. No L^2 intertwiner and no trace-class statement holds for the composite. The unrestricted* (H_I) *fails, exactly as the v4 record states.*

(ii) Mellin value. *The Mellin transform $M(z) = \int_0^\infty r^{1+z}\mathcal{R}(s;r)\,\mathrm{d}r$ converges for $\Re z > 2\nu$. It continues meromorphically, with poles contained in $(2\nu - 2\mathbb{N}_0) \cup (-2\nu - 2\mathbb{N}_0)$. So it is regular at $z = 0$ for $0 < \nu < 1$, and*

$$M(0) = 0 \qquad \textit{identically in } \nu \in (0,1).$$

Proof. Put $s = 1/2$, so that $x = r^2$; by scale invariance $M(0)$ scales as s^{-1}. Then

$$\mathcal{R} = A\, e^{-x}\Big[xK - 2xK' + K' + xK'' - 2\nu(K' - K) + \nu^2 x^{-1} K\Big], \qquad K = K_\nu(x),$$

where $A = \sin(\pi\nu)/(\pi s)$. Use $K' = -\frac{1}{2}(K_{\nu-1} + K_{\nu+1})$ and $K'' = \frac{1}{4}(K_{\nu-2} + 2K_\nu + K_{\nu+2})$. Each term then has the form $\int_0^\infty x^{\mu-1} e^{-x} K_m(x)\,\mathrm{d}x$ with

$$\int_0^\infty x^{\mu-1} e^{-x} K_m(x)\,\mathrm{d}x = \frac{\sqrt{\pi}\,\Gamma(\mu+m)\Gamma(\mu-m)}{2^\mu\,\Gamma(\mu+\frac{1}{2})}, \qquad \Re\mu > |\Re m|.$$

This is the $a \to 1$ limit of DLMF 10.43.22 [25]. [exec] It is checked by quadrature in its strip to 10^{-16} (`logs/v6/hi_d01_exact.json`). The right side is meromorphic in μ, which gives the stated continuation and poles. Leading-order expansion of K_ν gives (i).

At $z = 0$ the bracket is a sum of ten gamma ratios with $\mu \in \{0,1,2\}$ and $m \in \{\nu, \nu \pm 1, \nu \pm 2\}$. [exec] Applying $\Gamma(w+1) = w\Gamma(w)$ reduces it to 0 as a symbolic function of ν (`code/hi_d01_exact.py`). The sum at $z = 10^{-30}$ is $O(10^{-30})$ for seven values of ν.

An independent route uses direct quadrature with the $r^{-2\nu-2}$ and $r^{-2\nu}$ heads subtracted and continued analytically (`code/hi_d01.py`, 60 digits). It gives $s\,M(0) \in [-1.8\times10^{-21}, 3.7\times10^{-32}]$ at $\alpha = 0.6, 0.7, 0.75, 0.8, 0.9$, and the value is independent of s. The residual grows like $\epsilon^{4-2\nu}$ for the cutoff $\epsilon = 10^{-9}$, consistent with zero. □

Corollary 6.5 (Status of ($\mathrm{H}_{I,\mathrm{D01}}$))**.** *Interpret the apex coefficient of the forbidden D01 composite by homogeneous (Mellin) continuation. This is the unique scale-covariant extension of the non-integrable diagonal here, since there is no resonance for $0 < \nu < 1$. With that reading the composite contributes 0, and the D01 constant-profile entries of Table 2 need no correction from it.*

The hypothesis is therefore upgraded *in two respects: its failure as an L^2 statement is now exact, and the Mellin value is exactly zero. One thing is not shown: that the XVI-P prescription requires homogeneous continuation of this composite, as opposed to some other extension of the divergent apex integral. That is a prescription statement, part of (R3)–(R5), and it is recorded as* CONDITIONAL.

6.5 Longitudinal UV normalization

Set

$$\varepsilon = 4 - d, \qquad D = d - 2 = 2 - \varepsilon.$$

This ε is twice the common convention $\varepsilon_{\mathrm{DR}} = (4-d)/2$.

Lemma 6.6 (Angular extraction of the logarithmic UV residue)**.** *Suppose an infrared-regular Euclidean longitudinal integrand has large-radius expansion*

$$I(R\omega; p) = R^{-2} g(\theta; p) + O(R^{-2-\delta}), \qquad \delta > 0,$$

uniformly in angle, with no other UV-logarithmic term. If its dimensional continuation is regular in angle at $D = 2$, the coefficient of $1/\varepsilon$ is

$$\frac{1}{(2\pi)^2} \int_0^{2\pi} g(\theta; p)\, \mathrm{d}\theta. \tag{6.2}$$

An overall Wick phase $+\mathrm{i}$, when prescribed by the amplitude convention, multiplies this coefficient by i.

Proof. Choose a fixed large radius R_0 and separate the bounded infrared region. The leading UV radial integral is

$$\int_{R_0}^{\infty} R^{D-1} R^{-2}\, \mathrm{d}R = \int_{R_0}^{\infty} R^{-1-\varepsilon}\, \mathrm{d}R = \frac{R_0^{-\varepsilon}}{\varepsilon}.$$

The remainder is UV integrable near $\varepsilon = 0$. The measure at $D = 2$ contributes $(2\pi)^{-2}\mathrm{d}\theta$. An $O(\varepsilon)$ correction to the angular continuation affects the finite part of a simple pole, not this leading coefficient. □

The factor in (6.2) agrees with the v4 expression $\mathrm{i}(4\pi^2)^{-1} \int g$ when its $+\mathrm{i}$ Wick convention is applicable. Graph-specific additional phases are not fixed by this radial argument.

Dimensional reduction requires a consistent distinction between momentum and vertex-index spaces; it does not automatically establish a Ward identity for the present boundary problem [28].

7 The retained constant-profile pole catalogue

7.1 Status and notation

Write $\widehat{B}_G$ for the candidate coefficient of $1/(4-d)$ printed in v4. The hat distinguishes a supplied graph record from an independently derived loop amplitude.

All formulas below are exact algebraic expressions. [exec] In v6 every entry is checked against the executed spectral logs of the frozen vertex arrays (`code/catalogue_check.py`, `logs/v6/catalogue_check.json`). Every graph, branch, and key agrees exactly, the entries absent from the logs are computed zeros, and the new entries are marked. Their interpretation as complete graph residues remains subject to Assumption 6.1.

For either branch, put

$$c_2 = \begin{cases} c_2^{\mathrm{F}}, & \mathrm{F}, \\ c_2^{\mathrm{CR}}, & \mathrm{CR}, \end{cases} \qquad A = c_0 + 2c_1 + 4c_2, \quad B = -c_0 + 2c_1. \tag{7.1}$$

The c's are the per-channel quantities (4.2); any multiplicity is already included in the coefficients of the following tables.

Graph	Retained candidate residue
D04	$-\dfrac{\kappa^2}{4\pi}c_0P^2$
D05a	$-\dfrac{\kappa^2}{4\pi}(c_2-c_0)P^2$
D04+D05a	$-\dfrac{\kappa^2}{4\pi}c_2P^2$

Table 1: Equivalent coefficient form of the v4 constant-profile $\phi\phi$ catalogue.

7.2 Scalar two-point sector

Expansion gives exactly

$$\begin{aligned}
\widehat{B}_{\mathrm{D04}} &= \frac{\kappa^2(\alpha^2-1)}{48\pi\alpha}P^2,\\
\widehat{B}^{\mathrm{CR}}_{\mathrm{D05a}} &= -\frac{\kappa^2(2\alpha-1)}{4\pi}P^2,\\
\widehat{B}^{\mathrm{F}}_{\mathrm{D05a}} &= -\frac{\kappa^2(\alpha-1)(2\alpha-1)}{4\pi\alpha}P^2,\\
\widehat{B}^{\mathrm{CR}}_{\mathrm{D04+D05a}} &= \frac{\kappa^2(-23\alpha^2+12\alpha-1)}{48\pi\alpha}P^2,\\
\widehat{B}^{\mathrm{F}}_{\mathrm{D04+D05a}} &= \frac{\kappa^2(\alpha-1)(13-23\alpha)}{48\pi\alpha}P^2.
\end{aligned} \tag{7.2}$$

The one-sided CR limit of the sum as $\alpha\uparrow 1$ is $-\kappa^2P^2/(4\pi)$. It is the polynomial remnant discussed in Remark 4.3, not a claimed flat-endpoint operator value.

[exec] The D05a symmetry factor $S_f=1/2$ is read directly from the regenerated Q02 graph record `xviq_regen/Q02_bulk_graphs.json` [4]. The same record gives 1/2 for D12 and 1 for D04. With the XVI-P one-propagator tadpole rule, the engine prefactor is $-\kappa^2/2$, and the displayed D05a values follow without a fitted constant.

7.3 Tensor two-point sector

For the next table, factor out $\kappa^2/(16\pi)$. Indices denote the independent-coordinate entries listed in Section 2.

Table 2: The hh bubble entries of v4, in equivalent apex-coefficient form, with the $(5,5)$ entries added from the executed record (v6). Every displayed entry is multiplied by $\kappa^2/(16\pi)$. D01 retains the graph-local intertwining qualification.

Graph and key	Coefficient after removing $\kappa^2/(16\pi)$
D11: $(4,4),(4,7),(7,7)$	$\dfrac{1}{2}c_0P^2$
D03: $(0,0),(0,9),(9,9)$	$-c_0P^2$
D03: $(3,3)$	$4c_0P^2$
D03: $(0,4)$	$c_0(q_0^2-3q_3^2)$
D03: $(3,4)$	$4c_0q_0q_3$
D03: $(4,9)$	$-c_0(3q_0^2-q_3^2)$

Graph and key	Coefficient after removing $\kappa^2/(16\pi)$
D03: $(4,4)$	$-(c_0+4c_1)P^2$
D03: $(4,7)$	$-c_0P^2$
D03: $(5,5)$ *(new in v6)*	$-8c_1P^2$
D01: $(0,0),(0,9),(9,9)$	c_0P^2
D01: $(3,3)$	$-4c_0P^2$
D01: $(4,4)$	$(c_0-4c_1+c_2)P^2$
D01: $(4,7)$	$(c_0+c_2)P^2$
D01: $(3,4)$	$-4(c_0+2c_2)q_0q_3$
D01: $(4,9)$	$Aq_0^2-Bq_3^2$
D01: $(0,4)$	$-Bq_0^2+Aq_3^2$
D01: $(5,5)$ *(new in v6)*	$-8c_1P^2$

The entries related by $1 \leftrightarrow 2$ are identified in the executed record: $B_{44}=B_{77}$, $B_{04}=B_{07}$, $B_{34}=B_{37}$, and $B_{49}=B_{79}$. This does not make B_{44} equal to B_{47} in D01 or D03.

7.4 Tadpoles D02 and D12 and the (12) entries (new in v6)

[exec] v4 did not print D02 and D12 as components. The executed record gives them in full. Put

$$W=\tfrac{1}{2}(c_0+2c_1+c_2),\qquad W^{\mathrm{CR}}=\frac{(2\alpha-1)(4\alpha-1)}{6\alpha},\qquad W^{\mathrm{F}}=\frac{2(\alpha-1)(2\alpha-1)}{3\alpha}.$$

The only nonzero D02 entries, together with their transposes, are

$$\begin{aligned}\widehat{B}_{\mathrm{D02}}/(\tfrac{\kappa^2}{\pi}W):\quad &(01)(01),\ (02)(02):\ q_3^2;\qquad (13)(13),\ (23)(23):\ q_0^2;\\ &(12)(12):\ P^2;\qquad (11)(22):\ -\tfrac{1}{2}P^2;\\ &(01)(13),\ (02)(23):\ -q_0q_3;\qquad (03)(11),\ (03)(22):\ q_0q_3;\\ &(00)(11),\ (00)(22):\ -\tfrac{1}{2}q_3^2;\qquad (11)(33),\ (22)(33):\ -\tfrac{1}{2}q_0^2.\end{aligned}$$

This makes 23 nonzero entries. They agree with the v3 tadpole comparison recorded in `CHECKPOINT_v4.md`. Moreover

$$\widehat{B}_{\mathrm{D12}}=0$$

identically on both branches.

The table above omitted one key present in the executed record, the (12)(12) entry:

$$\widehat{B}^{(12)(12)}_{\mathrm{D01}}=\widehat{B}^{(12)(12)}_{\mathrm{D03}}=-8c_1P^2\,\frac{\kappa^2}{16\pi}=-\frac{\kappa^2(\alpha-1)(5\alpha-1)}{24\pi\alpha}P^2,$$

on both branches. It vanishes at $\alpha=1$ and is continuous at $\alpha=1/2$, so Proposition 8.1 extends to it.

Mixed helicity. [exec] The reduction attaches one coefficient c_S to a monomial only when both lines sit in the same channel. Mixed-channel monomials are logged and dropped. They occur only on *vector-type* keys, those with a single transverse index on a leg:

- D01: (01)(01), (02)(02), (13)(13), (23)(23), (01)(13), (02)(23), and their transposes;
- D06: ghost $11,22$;

- D13: $(01)c^1$, $(02)c^2$, $(13)c^1$, $(23)c^2$.

No lattice key $\{00, 03, 33, 11, 22, 12\}$ has one. The vector-type entries of D01, D06, and D13 are therefore *excluded* from the record and *not* zero. They need mixed-channel traces, which are OPEN.

7.5 Ghost and source-ghost sector

Define

$$R = 3c_0 + 4c_1, \qquad T = 4c_0 + 3c_1, \qquad V = c_1 - c_0. \tag{7.3}$$

Then the v4 ghost self-energy is

$$\widehat{B}_{\mathrm{D06}} = \frac{\mathrm{i}\kappa^2}{8\pi} RP^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{7.4}$$

where row and column labels are $0, 3$.

The source-ghost entries are

$$\widehat{B}_{\mathrm{D13}} = \frac{\mathrm{i}\kappa^2}{16\pi} \begin{pmatrix} -2Tq_0 & 2Vq_3 \\ -Rq_3 & Rq_0 \\ -2Vq_0 & 2Tq_3 \end{pmatrix}, \tag{7.5}$$

with tensor-source row labels $0, 3, 9$ and ghost column labels $0, 3$.

For explicit comparison,

$$\begin{aligned} R &= \frac{(\alpha-1)(17\alpha-7)}{12\alpha}, \\ T &= \frac{(\alpha-1)(11\alpha-7)}{12\alpha}, \qquad V = \frac{\alpha-1}{2}. \end{aligned} \tag{7.6}$$

Equations (7.4)–(7.6) reproduce all entries of the v4 ghost table without changing any factor of i, π, or two.

The polynomials are branch independent because c_2 does not occur. This is an algebraic fact about the printed catalogue. It does not prove that every domain manipulation of the corresponding F and CR graphs is identical.

8 Exact checks available from the printed catalogue

Proposition 8.1 (Algebraic preservation and endpoint checks)**.** *The coefficient tables above are equivalent to the displayed rational expressions of the supplied v4 source. Within their stated sector:*

(i) *every printed F entry vanishes at $\alpha = 1$;*

(ii) *every printed CR entry equals its F partner at $\alpha = 1/2$;*

(iii) *all printed entries are continuous at $\alpha = 1/2, 2/3, 4/5$;*

(iv) *D04+D05a has the sum (7.2);*

(v) *the printed D06/D13 entries are independent of the weight-two branch choice.*

These checks do not certify absent vertex arrays.

Proof. The endpoint claims follow because $c_0 = c_1 = c_2^{\mathrm{F}} = 0$ at $\alpha = 1$, whereas $c_2^{\mathrm{F}} = c_2^{\mathrm{CR}}$ at $\alpha = 1/2$. All formulas are rational with no denominator zero in the stated sector.

For equivalence, substitute (4.2) into Tables 1 and 2. For instance,

$$\begin{aligned} c_0 - 4c_1 + c_2^{\mathrm{CR}} &= \frac{2(\alpha^2+6\alpha-1)}{12\alpha}, \\ c_0 - 4c_1 + c_2^{\mathrm{F}} &= \frac{2(\alpha-1)(\alpha-5)}{12\alpha}, \\ c_0 + c_2^{\mathrm{CR}} &= \frac{2(11\alpha^2-6\alpha+1)}{12\alpha}, \\ c_0 + c_2^{\mathrm{F}} &= \frac{2(\alpha-1)(11\alpha-7)}{12\alpha}, \\ A^{\mathrm{CR}} &= \frac{101\alpha^2-60\alpha+7}{12\alpha}, \\ A^{\mathrm{F}} &= \frac{(\alpha-1)(101\alpha-55)}{12\alpha}, \\ B &= \frac{11\alpha^2-12\alpha+1}{12\alpha}. \end{aligned}$$

The common factor $\kappa^2/(16\pi)$ gives the v4 D01 denominators and numerators.

Similarly,

$$c_0 + 4c_1 = \frac{(\alpha-1)(19\alpha-5)}{12\alpha},$$

which checks the nontrivial D03 entry. The scalar sum is immediate from Table 1. Finally (7.6) proves the ghost-table equivalence and its branch independence. □

8.1 A restricted branch-transversality check

The following statement concerns the printed independent-coordinate array, not an unproved identification with a geometric Hessian.

Proposition 8.2 (Gauge annihilation of the printed D01 branch difference)**.** *Order the coordinates as*

$$x = (h_{00}, h_{03}, h_{33}, h_{11}, h_{22}), \qquad x_T = h_{11} + h_{22},$$

and let

$$u = (4q_3^2, -8q_0q_3, 4q_0^2).$$

With $\nu = 2 - \alpha^{-1}$, the CR-minus-F difference in Table 2 is the bilinear form

$$\begin{aligned} \Delta\widehat{B}_{\mathrm{D01}}(x,z) = \frac{\kappa^2\nu}{16\pi}\big[&(u\cdot x_L)z_T + x_T(u\cdot z_L) \\ &+ P^2 x_T z_T\big], \end{aligned} \tag{8.1}$$

where $x_L = (h_{00}, h_{03}, h_{33})$. It annihilates the declared longitudinal gauge vector

$$k(c) = (2q_0c_0,\ q_0c_3 + q_3c_0,\ 2q_3c_3,\ 0,\ 0) \tag{8.2}$$

in either slot.

Proof. Only c_2 changes, by ν. Reading its coefficients from Table 2 gives (8.1). Since $k_T = 0$,

$$u\cdot k_L = 8q_0q_3^2c_0 - 8q_0q_3(q_0c_3 + q_3c_0) + 8q_0^2q_3c_3 = 0.$$

Both slot contractions vanish. □

Independent-component weights matter here. One must not apply an additional off-diagonal multiplicity to an array already differentiated with respect to independent h_{03} coordinates. v6 adds the D02/D12 contribution in Corollary 11.2. In the complete hh sum the S_6, S_7 parts of this difference cancel against D02, which leaves the gauge-invariant term $-\nu\kappa^2P^2f_0f_0'/(4\pi)$.

The mixed-helicity statement of v4 is now an executed check; its precise form is given in Section 7.

9 The ghost-equation row and its exact scope

9.1 An explicit longitudinal contraction

For this algebraic check choose the covector

$$p_\mu = (q_0, 0, 0, q_3), \qquad p^\mu = (q_0, 0, 0, -q_3).$$

This fixes the spatial momentum sign rather than leaving the word "upper" ambiguous.

For the three displayed source rows of a symmetric tensor Y, define

$$\begin{aligned}(\mathsf{F}_L Y)_0 &= \frac{q_0}{2}(Y_{00} + Y_{33}) - q_3 Y_{03},\\ (\mathsf{F}_L Y)_3 &= q_0 Y_{03} - \frac{q_3}{2}(Y_{00} + Y_{33}).\end{aligned} \tag{9.1}$$

These are the de Donder contractions when the transverse trace of Y vanishes.

For a general tensor, the full contraction

$$(\mathsf{F}Y)_\rho = p^\mu Y_{\mu\rho} - \frac{1}{2} p_\rho \eta^{\mu\nu} Y_{\mu\nu}$$

instead has the additional terms

$$(\mathsf{F}Y)_0 = (\mathsf{F}_L Y)_0 + \frac{q_0}{2}(Y_{11} + Y_{22}), \qquad (\mathsf{F}Y)_3 = (\mathsf{F}_L Y)_3 + \frac{q_3}{2}(Y_{11} + Y_{22}). \tag{9.2}$$

The omitted transverse source trace is therefore a real identification obligation, not a value that may be silently set to zero.

Theorem 9.1 (Exact projected ghost-equation identity)**.** *For the printed matrices* (7.4) *and* (7.5)*,*

$$\widehat{B}_{\mathrm{D06}} = -2\eta_{\parallel}\, \mathsf{F}_L \widehat{B}_{\mathrm{D13}}, \qquad \eta_{\parallel} = \mathrm{diag}(1, -1). \tag{9.3}$$

The equality is an exact polynomial identity for every $0 < \alpha \le 1$ for which c_0, c_1 are given by (4.2)*. It is independent of c_2.*

To identify (9.3) *with the full graph antighost equation additionally requires the complete source normalization, the matched Fourier convention, and either a zero transverse D13 trace or its explicitly included contribution in* (9.2)*.*

Proof. The identity

$$T + V = R$$

follows from (7.3). Applying F_L to (7.5) gives

$$\mathsf{F}_L \widehat{B}_{\mathrm{D13}} = \frac{\mathrm{i}\kappa^2}{16\pi} R P^2 \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Multiplication by $-2\eta_{\parallel}$ yields (7.4). Formula (9.2) states exactly the additional transverse term needed for the full identification. □

This proves the two ratios reported in v4 on the printed longitudinal projection: $-1/2$ in the 00 contraction and $+1/2$ in the 33 contraction, with zero off-diagonal entries. It is not a proof that all source components or all 26 Ward-constrained combinations have been tested.

Remark 9.2 (Transverse source trace (upgraded in v6)). [exec] The executed D13 reduction covers all 10×4 source–ghost keys. The rows (11) and (22) vanish for the ghost components c^0, c^3 on both branches. On the executed matrix $\mathsf{F} = \mathsf{F}_L$, so the additional term in (9.2) is computed to be zero, not assumed zero. The ghost row therefore holds for the full de Donder contraction, components $\rho \in \{0, 3\}$, on constant profiles. The source normalization and the Fourier convention are those of the archived code. Their identification with the XVI-P antighost equation follows (C1)–(C3) of Section 10.3.

9.2 The Maxwell–Friedrichs comparison

The free photon inverse compatible with the scalar Friedrichs ghost is the Maxwell–Friedrichs construction described in the supplied I-B-G-S excerpt:

$$C_1 G^{\pm}_{(1)} = G^{\pm}_{(0)} C_1.$$

With Friedrichs realizations in the vector and scalar ghost channels, the supplied statement gives the Lorenz condition for conserved compact sources and locally square-integrable packet field strength [9, `thm:spin1`].

Residual gauge parameters there are scalar Friedrichs solutions, in a stated solution or spectral-packet domain, with gradients in the vector domain. They are not restricted to nonzero spacetime-compact homogeneous solutions.

This is a free Maxwell comparison, not the interacting diffeomorphism-ghost identity of D06/D13. Its availability supplies neither the missing transverse source rows nor a loop-level Slavnov–Taylor theorem.

Separation boundary. The cone, extension, regulator, and external-field class are prescribed. A free inverse, operator limit, one-point finite part, finite instrument, or documentary comparison is not an interacting two-point Ward theorem. No Einstein dynamics, physical tensor Hessian, spacetime emergence, or inherited quantum admission follows.

10 The hh Slavnov–Taylor row: v4 diagnostic, diagnosis, and constant-profile closure

10.1 The supplied attempted row (v4, historical)

The v4 computation reports a test of

$$2B_{hh}(K_2 c, h) + \lambda\, \Gamma_{Kc}(Eh, c) = 0, \tag{10.1}$$

using

$$B_{hh} = \mathrm{D01} + \mathrm{D02} + \mathrm{D03} + \mathrm{D11} + \mathrm{D12},$$

a linearized Einstein operator E, and a convention constant λ.

According to that record:

- no nonzero α-independent convention choice gave the desired CR/F closure;
- the UNIT replacement $c_S = 1$ gave $B_{hh}(K_2 c, h) = 0$ but a nonzero source term;
- 72 scanned conventions did not repair the test.

In v5 the code, arrays, and scan log were unavailable. v6 archives them (`code/st_test_const.py`, `code/st_scan.py`, `logs/v4/st_*.json`) and diagnoses the failure in Section 10.3.

10.2 Why UNIT requires its own analytic control theorem

The original inference that UNIT *must* close merely because it resembles the $l_\perp = 0$ flat integrand is too strong. Restriction to a momentum slice need not preserve an identity that holds only after integration in all momentum components.

Example 10.1 (A full-space total derivative need not vanish on a slice)**.** Let $F(l, k)$ be smooth and rapidly decreasing in all variables. Then

$$\int \mathrm{d}l\, \mathrm{d}^2 k\, \partial_{k_1}(k_1 F(l, k)) = 0.$$

But at $k = 0$,

$$\partial_{k_1}\left(k_1 F(l,k)\right)\big|_{k=0} = F(l,0),$$

whose longitudinal integral need not vanish. Thus full-momentum integration and transverse restriction do not commute with this cancellation.

A valid UNIT control must therefore derive the intended row for the actual reduced theory, source insertions, independent-component normalization, regulator, and momentum-routing operations. Longitudinal shift invariance alone does not establish that theorem. Dimensional-reduction consistency is likewise not automatic preservation of every proposed identity [28].

Proposition 10.2 (No regularity verdict follows from the v4 test)**.** *The supplied failure of* (10.1)*, including its UNIT diagnostic, proves neither failure nor validity of* ($\mathrm{H}_{\mathrm{reg}}$)*. It does not test the complete set of 26 Ward-constrained combinations.*

Proof. The complete row, its source and auxiliary-field terms, its normalization, and a validated reduced control are not established in the supplied test. Several distinct missing choices can change its residual: invariant versus gauge-fixed E, auxiliary and antighost contributions, source momentum routing, transverse source traces, and independent-component multiplicities.

An unexplained residual of that unvalidated map cannot be identified with the residual of the required full identity. Conversely, failure of one candidate test does not establish the required identity either. □

The ending of v5 for the *v4 row* remains correct: that row was UNVALIDATED. It was not "cone anomaly found," and it was not "BRST restored." v6 replaces it with a derived row, analysed below.

10.3 Diagnosis of the v4 failure (new in v6)

The frozen XVI-P antifield vertex has the closed form $K'_{(mn)}\, a\,(\partial_m c_n + \partial_n c_m)$, summed over the independent coordinates (mn), $m \le n$ [3]. Writing the BRST source term in the same independent coordinates fixes three conventions that v4 left free:

(C1) the ghost enters the vertex with an upper index, and $c_n = \eta_{nn} c^n$ appears in $\delta_c h_{(mn)} = p_m c_n + p_n c_m$;

(C2) the equation-of-motion factor is the derivative with respect to the independent coordinate,

$$\frac{\partial S}{\partial h_{(mn)}} = (2 - \delta_{mn}) E^{mn}(h),$$

with E the linearized Einstein operator and indices raised by $\eta = \mathrm{diag}(1,-1,-1,-1)$;

(C3) consequently the $K'_{(03)}$ coordinate carries weight $2E^{03} = -2E_{03}$. In lower-index components this is weight $w = -2$.

[exec] The v4 convention scan (`code/st_scan.py`, `logs/v4/st_scan.json`) used only positive weights $w \in \{\frac{1}{2}, 1, 2\}$. It never contained $w = -2$ on (03) together with (C1). This is why none of its 72 conventions closed. The UNIT "nonzero source term" in v4 is produced by the same weight error.

10.4 An *a priori* normalization

v5 rightly warned that a fitted λ proves nothing. In v6, λ is fixed before the loop data are used. The XVI-P tree data give the antighost–ghost normalization $A_{\mathrm{GH}} = \frac{1}{2}$. They also

give the quadratic-action normalization $N = 1$ of E^{up} on de Donder configurations. [exec] `code/st_norm.py` solves the de Donder constraint for (h_{00}, h_{03}), with $h_{11} = h_{22}$ tied, and returns

$$\lambda_{\mathrm{pred}} = \frac{N}{\mathrm{i}A_{\mathrm{GH}}} = -2\mathrm{i}$$

(`logs/v6/st_norm.json`).

10.5 Constant-profile closure on CR and F

Let the axisymmetric lattice coordinates be $x = (h_{00}, h_{03}, h_{33}, f_0)$, with $h_{11} = h_{22} = f_0$, and the ghost $c = (c^0, c^3)$. Put $p_\mu = (q_0, 0, 0, q_3)$. Define

$$\mathcal{S}_\lambda(c, h) = \sum_{a,b} \widehat{B}^{hh}_{ab}\,(\delta_c h)_a\, h_b + \lambda \sum_{(mn),\rho} \widehat{\Gamma}^{\mathrm{D13}}_{(mn),\rho}\, c^\rho\,(2 - \delta_{mn}) E^{mn}(h), \tag{10.2}$$

where $\widehat{B}^{hh} = \mathrm{D01} + \mathrm{D02} + \mathrm{D03} + \mathrm{D11} + \mathrm{D12}$ denotes the executed constant-profile poles restricted to lattice keys.

Theorem 10.3 (Constant-profile hh/Kc Slavnov–Taylor row)**.** *[exec] For both CR and F and every $1/2 < \alpha < 1$, the polynomial $\mathcal{S}_\lambda$ in (h, f_0, c, q_0, q_3) vanishes identically if and only if $\lambda = -2\mathrm{i} = \lambda_{\mathrm{pred}}$. More precisely:*

(i) *Sixteen monomials receive a nonzero graph contribution. On twelve of them, the D01 and D03 contributions cancel among the hh graphs. D02, D11, and D12 contribute nothing to the row.*

(ii) *The remaining four monomials are $f_0 c^0 q_0^3$, $f_0 c^0 q_0 q_3^2$, $f_0 c^3 q_0^2 q_3$, $f_0 c^3 q_3^3$. On them, the hh gauge variation has coefficients $\mp\kappa^2(\alpha - 1)/(4\pi) = \mp\kappa^2 V/(2\pi)$, and the antifield term has coefficients $\pm\mathrm{i}\kappa^2(\alpha - 1)/(8\pi)$. Hence a single α-independent $\lambda = -2\mathrm{i}$ cancels all four.*

(iii) *The matter sub-row* $\mathrm{D11} + \mathrm{D12}$ *is transverse by itself.*

(iv) *The UNIT replacement $c_S \equiv 1$ makes both terms vanish identically, a trivial closure. So does the flat four-dimensional benchmark on the lattice (`code/flat_pole.py`). Neither of these can determine λ, which is why λ is fixed* a priori.

Evidence. `code/st_v6.py` forms (10.2) symbolically from the logged poles and extracts every monomial coefficient as an exact rational function of α. It solves for λ on the first equation and checks $\partial_\alpha \lambda = 0$ and every residual $= 0$ (`logs/v6/st_v6_\{CR,F,UNIT,FLAT\}.json`). The per-graph table is in `logs/v6/st_v6_stats.json`.

Section 11 gives the structural reason. On the lattice, the gauge variation of the hh sum is $2(b_6 + b_7)P^2 f_0\,(p \cdot c)$, so $\mathcal{S}_\lambda \propto V$ on both branches. The branch-dependent structure b_8 is gauge invariant.

The validity of the flat tensor-averaging routine is checked separately. The flat D11 covariant ratios $[6, 1, -6, -1, 8]$ reproduce the frozen `P18_D11` record (`logs/v6/flat_vs_p18.json`). □

Remark 10.4 (Scope of the closure). Theorem 10.3 answers v5's slice objection (Example 10.1) for this row by direct computation, not by appeal to shift invariance. It is *one* row, under these restrictions:

- the constant profile;
- the axisymmetric lattice $h_{11} = h_{22}$, without the (12) entry;
- excluding the vector-type keys on which mixed-helicity monomials were dropped (Section 7);
- without the auxiliary-field terms that vanish on this restriction;

- and with the spectral reduction itself conditional on Assumption 6.1.

It is not the 26 Ward-constrained combinations, and it is not a full Slavnov–Taylor theorem on the cone. Within that scope the honest status changes from UNVALIDATED to *closed*. No anomaly is found, and no BRST restoration is claimed.

11 Invariant decomposition of the constant-profile hh poles (new in v6)

On the axisymmetric lattice, write h for the longitudinal 2×2 block, $\operatorname{tr} h = \eta^{ab}h_{ab}$, $h\cdot h' = h^{ab}h'_{ab}$, and $php = p^a p^b h_{ab}$ for $a,b \in \{0,3\}$. For two-dimensional symmetric matrices the Cayley–Hamilton identity removes $p\,hh'\,p$. A symmetric $SO(1,1)$-invariant bilinear that is quadratic in p is therefore a combination of six structures:

$$S_1 = P^2 \operatorname{tr} h \operatorname{tr} h', \qquad S_2 = P^2\, h\cdot h', \qquad S_3 = (php)\operatorname{tr} h' + \operatorname{tr} h\,(ph'p),$$
$$S_6 = P^2(f_0 \operatorname{tr} h' + \operatorname{tr} h\, f_0'), \qquad S_7 = (php)f_0' + f_0(ph'p), \qquad S_8 = P^2 f_0 f_0'.$$

Proposition 11.1 (Structure coefficients)**.** *[exec] Every executed constant-profile hh pole decomposes uniquely as $\sum_i b_i S_i$. In units of κ^2/π, the coefficients are the following branch-uniform linear forms in (c_0, c_1, c_2):*

Graph	b_1	b_2	b_3	b_6	b_7	b_8
D01	$-\frac{1}{16}c_0$	$\frac{1}{8}c_0$	0	$-\frac{1}{8}c_0 - \frac{1}{4}c_1 - \frac{1}{2}c_2$	$\frac{1}{4}c_0 + \frac{1}{2}c_2$	$\frac{1}{4}c_0 - \frac{1}{2}c_1 + \frac{1}{4}c_2$
D02	0	0	0	W	$-W$	$-W$
D03	$\frac{1}{16}c_0$	$-\frac{1}{8}c_0$	0	$\frac{3}{8}c_0$	$-\frac{1}{4}c_0$	$-\frac{1}{4}c_0 - \frac{1}{2}c_1$
D11	0	0	0	0	0	$\frac{1}{8}c_0$
D12	0	0	0	0	0	0
Sum	0	0	0	$\frac{3}{4}(c_0 + c_1)$	$-\frac{1}{2}c_0 - c_1$	$-\frac{3}{8}c_0 - 2c_1 - \frac{1}{4}c_2$

Here $W = \frac{1}{2}(c_0 + 2c_1 + c_2)$. The CR fit determines the forms uniquely. On F, where $c_2^{\mathrm{F}} = 7c_0 + 6c_1$ makes the three coefficients linearly dependent, the same forms reproduce every entry (`logs/v6/hh_structures.json`). UNIT ($c \equiv 1$) gives the sum $(0, 0, 0, \frac{3}{2}, -\frac{3}{2}, -\frac{21}{8})$.

Corollary 11.2 (Gauge and branch content of the complete hh sum)**.**

(i) *The purely longitudinal structures cancel: $b_1 = b_2 = b_3 = 0$. This cancellation is between D01 and D03.*

(ii) *The only gauge-variant combination on the lattice is*

$$b_6 + b_7 = \tfrac{1}{4}(c_0 - c_1)\frac{\kappa^2}{\pi} = -\frac{\kappa^2 V}{4\pi} = -\frac{\kappa^2(\alpha - 1)}{8\pi},$$

which is branch independent. This is exactly the term balanced by the antifield row in Theorem 10.3.

(iii) *The CR–F difference of the complete hh sum is*

$$\Delta\widehat{B}^{hh} = -\frac{\nu\kappa^2}{4\pi} P^2 f_0 f_0'.$$

It is annihilated by the longitudinal gauge vector (8.2), since $k_T = 0$. The D01 difference of Proposition 8.2 has S_6, S_7 parts. They cancel against the D02 difference $\frac{\nu}{2}(S_6 - S_7 - S_8)\kappa^2/\pi$. This settles the v5 obligation that D02/D12 must be included in the branch-transversality check.

Remark 11.3 (Relation to the 18-dimensional engine map). Axisymmetric constant-profile data land in the six-dimensional space above. The programme's complete hh map is 18-dimensional [1, 2]; it is computed by the XVI-P engine in its own basis. That basis also includes transverse-derivative and non-axisymmetric structures, which constant profiles cannot see (Proposition 12.1). The convention map that sends the six coefficients above to their images in the engine's 18 coordinates is OPEN. So, therefore, are the components of the 18-dimensional vector outside the image. They are not set to zero.

12 What constant-profile data cannot determine

12.1 The full basis projection

The supplied programme counts a 34-dimensional apex basis, including 26 Ward-constrained combinations and six exact plus two $\phi\phi$ directions. Those counts are retained as supplied basis metadata; the basis itself is not constructed from the printed component table.

Proposition 12.1 (Constant-profile restriction is not injective)**.** *On any smooth external-jet class containing constant transverse jets and quadratic transverse jets, restriction of apex-local bilinear functionals to constant profiles need not be injective. Consequently, constant-profile values alone do not determine coefficients multiplying all transverse-derivative operators.*

Proof. Consider the symmetric bilinear functional

$$L(F,G) = \int_{\mathcal{W}} (F\Delta_\perp G + G\Delta_\perp F)\ \mathrm{d}^2y.$$

It vanishes when both fields are transversely constant near the apex. Choose F with constant transverse profile near zero and $G = r^2 g(y)$ there, with a smooth outer cutoff. Since $\Delta_\perp r^2 = 4$ on the two-dimensional cone away from the apex and the radial jet has that limit, $L(F,G)$ is nonzero for suitable longitudinal tests. Thus the restriction has a nontrivial kernel. □

This elementary example does not declare the allowed physical counterterm basis. It proves why the missing basis cannot be recovered solely by evaluating the displayed restriction.

The single computed scalar combination is the longitudinal $\delta_{\mathcal{W}}\phi\,\partial_\parallel^2\phi$-type value in (7.2), in the supplied convention. In v6 the second scalar combination, which belongs to a quadratic transverse profile, is *partial*. Its D05a part is computed to be 0, and its D04 part is open (Section 5).

The higher coefficient $\mathcal{D}_b$ that v4 quoted without derivation is now derived in Theorem 5.2. The trace-cyclicity argument of v4 is not used: $\mathcal{D}_b$ comes from its own normalized weighted heat trace, and its domain audit is the flip estimate of Proposition 5.3.

12.2 Joint-apex condition (J)

If Assumption 6.1 holds, the constant-profile apex UV pole is the longitudinal logarithm in Theorem 6.2. This is the precise conditional version of the spectral joint-apex assertion.

It is not an unconditional proof of (J) for all graphs or all external jets. General profiles require the corresponding weighted spectral coefficients, derivative-domain control, and graph contractions. For D01 the additional requirement ($\mathrm{H}_{I,\mathrm{D01}}$) is now resolved in the Mellin sense, and its prescription status is recorded in Corollary 6.5.

12.3 Regularity at special cone openings

The printed rational functions are continuous at

$$\alpha \in \left\{\frac{1}{2}, \frac{2}{3}, \frac{4}{5}\right\}.$$

This is proved in Proposition 8.1. Continuity of finitely many coefficients is not a uniform estimate for the operator domains or nonconstant jets required by (H_{reg}).

The v3 claim about $H^1_{(3)} = 0$ modulo (Span) is retained only as a version-qualified supplied assertion. The exact complex, span matrix, and proof are not present in v4, so it is not used to infer the absence of a local obstruction.

12.4 Evanescent data and finite linear certificates

Evanescent coefficients (item (d)): partial. Lemma 6.6 shows that an $O(\varepsilon)$ dependence of the angular continuation, or of the quasi-four-dimensional index algebra, leaves the simple-pole coefficient unchanged. So the four-dimensionally projected poles of Section 7 are stable at pole level. The spectral code follows the XVI-P DRED prescription, with quasi-d loop momenta and quasi-four-dimensional vertex indices.

The evanescent *coefficients* needed for finite parts and later identities require the $\tilde{g}$-leg (evanescent-index) vertex arrays. These are not part of the frozen XVI-P exports used here. They are OPEN. No evanescent coefficient is replaced by its four-dimensional value.

Exact (Span) **certificate (item (f)): open.** The XVI-P engine computes the counterterm span natively modulo a prime. An exact certificate requires either a port of the rank computation to $\mathbb{Q}(i)$ with fraction-free elimination, or a specified nonzero minor evaluated exactly. Neither an exact matrix export nor such a port was found in the frozen trees as inspected. Building one would mean modifying or extending the frozen engine, which this revision does not do. Memory is not the obstruction. A modular rank, even at several primes, is evidence and not an exact certificate, and it is not reported as one.

Remaining items. The following remain OPEN:

(i) the evanescent coefficients above;

(ii) the convention map of the six structure coefficients of Section 11 into the 18-dimensional engine basis, and the components outside its image;

(iii) the full 34-dimensional basis projection and the 26 Ward combinations;

(iv) the exact (Span) certificate;

(v) the vector-type keys excluded in Section 7.

A numerical rank or one chosen minor does not certify a whole unspecified contact family. The matrix, coefficient field, normalizations, and completeness of its columns must first be declared.

13 Companion comparisons and finite-report boundaries

13.1 Smooth cores, one-point data, and infrared control

The v4 intake reports two newly found registry identifiers: `I-B-VP-5-Ex1-B` and `I-B-VP-5-Ex2-B`. The historical count is relative to that reported intake snapshot, not a claim about the present registry.

The supplied Ex1-B material concerns non-universality of complete smooth-core finite parts and heat-normalization audits [14]. The supplied Ex2/Ex2-B material concerns twisted scalar Friedrichs selection, a Dirac critical-window selection, and operator-limit qualifications [15, 16]. Neither supplies an interacting spin-2 CR graph calculation.

The supplied I-B-VP material separately reports compact-test infrared control for the massless Friedrichs spectral covariance and regular-host one-point formulas [12, `lem:infrared`, `thm:`

friedrichs]. That infrared statement is not a proof that the present massless loop integrals have no infrared poles.

The Ward-compatible extension freedom for one-point stress and current distributions in I-B-VP-1 is another extension problem [13]. It does not replace the Maxwell–Friedrichs photon domain, choose the spin-2 CR/F realization, or fix a two-point graph prescription. Strong-resolvent convergence alone also does not imply convergence of singular covariances or renormalized one-point observables.

13.2 Continuum kernels and echo reports

The free Green identities in I-B-G are tied to their spectral test domain. They are not automatically an exact sequence on the ordinary compact-spacetime test space [8, thm:green, rem:compact-exact].

I-B-ECHO compares image-isolated windows, diffracted fronts, short-window deficit sensitivity, and finite-precision neutral-core reports [11]. Those results do not identify the window functional with the apex residue. Example 3.4 already separates the heat constant from a purely diffracted-front diagnostic.

If one later requests a finite report of a residue, three new objects must be specified:

defined continuum coefficient $\longrightarrow$ acquisition or supplied preparation $\longrightarrow$ complete finite instrument.

Only the first is the target of the present loop calculation. A finite code cannot create the missing coefficient or its Ward identity.

13.3 Finite Enter instruments and their ledgers

For comparison, the finite destructive instrument has form

$$\mathcal{M}(X) = \sum_j |j\rangle\langle j|_R \otimes D_j X D_j^\dagger, \qquad D_j = \langle \tau_j|_Q \otimes I_A,$$

with

$$\sum_j D_j^\dagger D_j = I, \qquad \mathrm{Tr}_R\, \mathcal{M}(X) = \mathrm{Tr}_Q\, X.$$

These identities follow by matrix-unit expansion and remain true with untouched finite references [6, 7]. They consume a finite name and return a classical code with its declared residual; they do not select the continuum domain.

For a supplied real signal s, X-J-ST uses laws of the form

$$P_s(F) = r, \qquad P_s(\pm) = \frac{1-r}{2}\left(1 \pm \tanh(\gamma s/\sigma)\right).$$

For two supplied values,

$$\mathrm{TV}(P_{s_1}, P_{s_2}) = \frac{1-r}{2}|\tanh(\gamma s_1/\sigma) - \tanh(\gamma s_2/\sigma)|.$$

Complete implementation allowances e_1, e_2 reduce a guaranteed ideal separation by at most $e_1 + e_2$, by the triangle inequality [17, 11].

Finite clock composition and resource fitting require the full history, failures, controls, locations, and storage ledgers [18]. These facts price an already declared finite interface. They neither acquire a continuum pole nor validate a failed ST test. Heat belongs to a separate physical implementation ledger [19, 31].

Separation boundary. The cone, extension, regulator, and external-field class are prescribed. A free inverse, operator limit, one-point finite part, finite instrument, or documentary comparison is not an interacting two-point Ward theorem. No Einstein dynamics, physical tensor Hessian, spacetime emergence, or inherited quantum admission follows.

14 Historical corrections and reproducibility obligations

14.1 Corrections retained from v4

The following v4 corrections to v3 remain:

(i) D06 is the $\bar{c}c$ ghost self-energy, not "the D01+D03 channel."

(ii) The source reports confirmation of the D05a symmetry factor 1/2 from Q02. Its displayed value is unchanged.

(iii) The closed constant-profile polynomials replace the earlier unevaluated Mellin-residue status only in that restricted reported computation. They do not supply nonconstant jets.

(iv) The registry log distinguishes the oct18c report of zero new identifiers from the later oct18d report of two. v6 rechecked it: 251 identifiers, unchanged.

14.2 Proof-level corrections made in v5 (retained)

(i) The apex functional is explicitly bulk-subtracted.

(ii) Nonperiodic CR labels are justified through a singular Bessel-channel replacement.

(iii) The flat CR remnant is identified as a one-sided limit.

(iv) The heat-trace formula and complete graph reduction have separate hypotheses.

(v) The projected ghost row has explicit momentum and transverse-trace conventions.

(vi) The UNIT benchmark is not assumed valid merely from longitudinal shift invariance.

(vii) The full hh tadpole arrays and higher-profile values are not reconstructed from "as v3" (v6: now executed and printed).

(viii) Self-adjoint Schrödinger evolution, positive wave evolution, and open-system access loss are distinguished in Appendix A.

14.3 Corrections in v6

(i) The $K'_{(03)}$ coordinate weight in the hh ST row is -2, not a positive weight (Section 10.3).

(ii) The ST constant is $\lambda = -2\mathrm{i}$ in the normalization (10.2). The early-v6 checkpoint value $-4\mathrm{i}$ belongs to the v4 normalization, which has an extra factor 2 on the B term. Both are recorded in `CHECKPOINT_v6.md`.

(iii) The v4/v5 catalogue omitted the $(5, 5)$ entries of D01 and D03. The executed record contains them, and they are added without changing any printed entry.

(iv) "Mixed-helicity cancellation" is made precise: absent on lattice keys, and excluded (not cancelled) on vector-type keys.

(v) The early numerical (H_I) runs suffered catastrophic cancellation in $I_{-\nu} - I_\nu$. They were replaced by the K_ν form, analytic derivatives, and an explicit small-r head. Only the corrected run is reported.

14.4 Artifact inventory (v6)

The executed record now consists of the following (Appendix C):

- the spectral reduction `code/spectral_apex.py`, with the graph rules copied read-only from XVI-P `amplitude_rules.py` and XVI-Q Q02;

- per-graph logs in `logs/v4/` for CR, F, and UNIT;
- the v6 programs `st_v6.py`, `st_norm.py`, `st_v6_stats.py`, `flat_pole.py`, `flat_vs_p18.py`, `weighted_apex.py`, `weighted_checks.py`, `hi_d01.py`, `hi_d01_exact.py`, `hh_structures.py`, and `catalogue_check.py`, with their logs in `logs/v6/`;
- full SHA-256 tree digests of XVI-P/Q/R, taken before and after this revision.

The `logs/v4/` spectral runs were executed on this machine during the v4 session. v6 re-reads and audits them, and reruns the derived checks. v6 does *not* rerun the spectral contractions themselves; their per-graph timings are recorded in the JSON logs. This replay-versus-rebuild distinction is recorded deliberately.

14.5 Status of the computational-certificate obligations

v5 obligation	v6 status
1. Frozen source versions and complete hash manifests	Done: full tree digests and full input-file digests.
2. Graph inventory, vertices, routing, phases, statistics, symmetry factors, normalizations	Done for D01–D06, D11–D13: archived in code, with values read from frozen sources.
3. Helicity projectors and domain-changing operations	Projectors archived. Domain: D01 composite exact (Theorem 6.4); other graphs per the v4 assessment.
4. Exact contracted spectral functions before the spectral parameter is set to zero	OPEN: the code reduces at zero transverse momentum.
5. Regularized trace reductions and boundary defects	Partial: Theorems 3.2, 4.1, 5.2, and 6.4.
6. UV/IR separation and regulator dependence	Partial: Lemma 6.6; (R4) open.
7. Full D02/D12 arrays and source transverse-trace rows	Done (Section 7).
8. Complete ST functional, derived control, raw residuals	Done for one row within scope (Theorem 10.3). Full functional open.
9. Exact basis-projection and rank certificates	OPEN.
10. Commands, versions, logs, failures, replay vs. rebuild	Done (Appendix C; corrections above).

Polynomial agreement and hash agreement alone still do not supply items 4, 6, and 9.

15 Point-by-point response to the v5 audit (new in v6)

The table answers every issue v5 raised against v4, in v5's order. "Retained" means v5's correction stands unchanged. [exec] marks an executed v6 answer (Appendix C).

#	v5 issue against v4	v6 answer
1	Apex functional must be bulk-subtracted.	Retained. Extended to the r^2-weighted trace (Definition 5.1, Theorem 5.2).
2	Nonperiodic CR labels need a singular-channel justification.	Retained (Theorem 4.1). Weighted analogue added (Proposition 5.3).
3	Flat CR remnant is only a one-sided limit.	Retained. New: the remnant is absent from the r^2 moment (Corollary 5.4).
4	Heat trace and graph reduction need separate hypotheses.	Retained (Assumption 6.1). (R1) is discharged by the archived arrays. (R3) is discharged for the D01 composite in the Mellin sense (Theorem 6.4). (R2), (R4), and (R5) remain hypotheses.

#	v5 issue against v4	v6 answer
5	Catalogue identification conditional; vertex arrays and execution artifacts not supplied.	[exec] Arrays, routing, and prefactors are archived (`code/spectral_apex.py`). All v5 entries match the logs exactly (`code/catalogue_check.py`). New entries: D01/D03 $(5, 5)$, the full D02 array, and D12 $= 0$ (Section 7).
6	Mixed-helicity cancellation is only a supplied report.	[exec] No mixed-helicity monomial occurs on any lattice key. Such monomials occur only on vector-type keys, those with one transverse index on a leg, in D01, D06, and D13 (listed in Section 7). Those keys are excluded, not set to zero.
7	D05a symmetry factor rests on a supplied Q02 comparison.	[exec] Read directly from the regenerated `xviq_regen/Q02_bulk_graphs.json`: D05a 1/2, D12 1/2, D04 1. The engine prefactor is $-\kappa^2/2$, and the D05a value is unchanged.
8	Ghost row needs explicit momentum conventions and the transverse source trace.	Conventions retained (Theorem 9.1). [exec] The D13 rows (11), (22) are computed to be zero for both ghost components. So $\mathsf{F} = \mathsf{F}_L$ on the executed matrix, and the transverse-trace obligation is discharged for constant profiles.
9	UNIT is not valid merely by longitudinal shift invariance (slice example).	Answered by direct computation instead of by that argument. UNIT closes trivially, and λ is fixed a priori (Theorem 10.3).
10	hh ST test is UNVALIDATED; no verdict.	Diagnosed: the $K'_{(03)}$ weight should be -2 (Section 10.3). The derived row closes on CR and F with $\lambda = -2\mathrm{i} = \lambda_{\mathrm{pred}}$, within the scope of Remark 10.4.
11	Full hh tadpole arrays not reconstructed from "as v3".	[exec] D02 is printed in full and D12 $= 0$. Both agree with the 24-coefficient v3 tadpole comparison recorded in `CHECKPOINT_v4.md`.
12	D01 branch-transversality check omits D02/D12.	Done: the complete hh branch difference is $-\nu\kappa^2 P^2 f_0 f_0'/(4\pi)$, which is gauge invariant (Corollary 11.2).
13	Independent-component weights must not be double counted.	Confirmed and used. The $(2 - \delta_{mn})$ weight is *derived*. The flat D11 comparison with `P18_D11` fixes off-diagonal weight 2 in the covariant basis.
14	26 Ward combinations; 34-dimensional projection.	OPEN. One row is tested and closes. The basis itself is still not constructed.
15	Six exact plus two scalar basis values.	One scalar value is retained. The second is partial: D05a part $= 0$, D04 part OPEN. The six exact values are OPEN.
16	$\mathcal{D}_b$ was quoted without derivation or value.	Derived and checked (Section 5).
17	(J) only conditional; $(\mathrm{H}_{I,\mathrm{D01}})$ open.	$(\mathrm{H}_{I,\mathrm{D01}})$: exact L^2 obstruction and an exactly zero Mellin value. The prescription choice is CONDITIONAL (Corollary 6.5). (J) for general profiles remains open.
18	$(\mathrm{H}_{\mathrm{reg}})$ at $\alpha = 1/2, 2/3, 4/5$.	Coefficients remain continuous, and so do $\mathcal{D}_b$ and $M(z)$ (no resonance for $0 < \nu < 1$). The general statement is still open.
19	$H^1_{(3)} = 0$ mod (Span) only version-qualified.	Unchanged; not used.
20	Evanescent coefficients.	Pole-level statement only (Section 12). The coefficients are OPEN.
21	Complete hh 18-dimensional map.	Partial: the six-structure decomposition is complete. The engine map is OPEN.
22	Exact (Span) certificate.	OPEN, with reason (Section 12).
23	Truncated hashes cannot authenticate trees.	Full tree digests before and after, and full file digests, are in Appendix C.

#	v5 issue against v4	v6 answer
24	Registry intake only historical.	Quick recheck: 251 identifiers, identical in set and content to oct18c∪oct18d. No change, so no `new_oct18e/` directory was created.
25	Ten-item reproducibility obligation list.	Status per item in Section 14.
26	Compilation not asserted.	Two pdfLaTeX passes with no unresolved references (Appendix B).
27	Readout analogy supplies no premise.	Retained unchanged (Appendix A).

16 Ledger and release

Item	Status in v6 (change from v5)
Normalized twisted Friedrichs apex constant	Proved: Theorem 3.2.
One exceptional regular-to-singular Bessel heat shift	Proved: Theorem 4.1.
r^2-weighted apex coefficient $\mathcal{D}_b$ and its shift	**New**, proved: Theorem 5.2 and Proposition 5.3, with orbifold checks.
Spectral reduction of a specified complete graph	Conditional on Assumption 6.1. (R1) discharged by archived arrays.
D01–D06, D11–D13 constant-profile expressions	**Upgraded**: executed and matching all v5 entries. $(5,5)$, D02, and D12 added. Graph identification still conditional on (R2)–(R5).
D01 exceptional derivative tower ($\mathrm{H}_{I,\mathrm{D01}}$)	**Upgraded**: exact L^2 obstruction; Mellin value $= 0$ exactly. Prescription CONDITIONAL.
F flat limit and CR=F at $\alpha = 1/2$	Proved for the catalogue, including the new entries.
Ghost-equation row	Exact identity proved. **Upgraded**: transverse trace computed to be zero.
D02/D05a/D12	**Upgraded**: D02 printed, D12 $= 0$, D05a factor read from Q02.
Mixed-helicity terms	**Upgraded**: absent on lattice keys; vector-type keys excluded.
hh Slavnov–Taylor row and UNIT	**Upgraded** from UNVALIDATED to *closed in scope*: CR and F with $\lambda = -2\mathrm{i} = \lambda_{\mathrm{pred}}$; UNIT trivially.
Complete *hh* branch difference	**New**: $-\nu\kappa^2 P^2 f_0 f_0'/(4\pi)$, gauge invariant.
26 Ward combinations; 34-dimensional projection	Not obtained.
Six exact values	Not obtained.
Second scalar value	**Partial**: D05a part $= 0$; D04 part open.
Joint-apex condition (J)	Constant profile only, with the D01 composite now controlled. General profiles open.
($\mathrm{H}_{\mathrm{reg}}$) at $1/2, 2/3, 4/5$	Coefficients continuous; general statement open.
Evanescent coefficients	OPEN (pole-level statement only).
Complete *hh* 18-dimensional map	**Partial**: six-structure decomposition; engine map open.
Exact (Span) certificate	OPEN.
Vector-type keys (one transverse index)	OPEN: they need mixed-channel traces.
Registry intake	Rechecked; unchanged.
Companion and catalogue connections	Separations, not new premises or admission.

Item	Status in v6 (change from v5)
O.1 on the cone	`NOT_CLOSED`.
`bulk_symmetry_restored`	`false`.

Theorem 16.1 (Exact release of v6)**.** *Under the printed definitions and hypotheses:*

(i) *the normalized apex constants $\mathcal{C}_b$ and the weighted coefficients $\mathcal{D}_b$ are established, together with their singular-channel shifts;*

(ii) *a complete graph satisfying the spectral reduction contract has the longitudinal residue* (6.1)*;*

(iii) *the executed constant-profile poles equal the coefficient catalogue of Section 7, with the algebraic checks proved above;*

(iv) *the projected ghost-equation row* (9.3) *holds, and the transverse source trace vanishes on the executed matrix;*

(v) *the constant-profile hh/Kc Slavnov–Taylor row closes on CR and F with the a-priori constant $\lambda = -2\mathrm{i}$;*

(vi) *the D01 composite has an exact L^2 obstruction and zero Mellin value;*

(vii) *the available data do not supply the full basis map, the six exact values, the D04 quadratic-profile part, the evanescent coefficients, the engine map, or an exact* (Span) *certificate;*

(viii) *neither the companion separations nor the finite-readout comparison changes the cone O.1 ending.*

No complete cone O.1 closure or bulk symmetry restoration is released.

Proof. Theorems 3.2, 4.1, and 5.2, with Proposition 5.3, give (i). Theorem 6.2 gives (ii). Proposition 8.1 and the executed catalogue check give (iii). Theorem 9.1 and the executed D13 rows give (iv). Theorem 10.3 gives (v), and Theorem 6.4 gives (vi). Propositions 10.2 and 12.1, Remark 10.4, and Section 12 give (vii).

O.1 on the cone requires the full Ward/basis projection on the prescribed cone, the evanescent and (Span) data, and (R2)–(R5) for every graph. These are listed as open, and none is supplied by (i)–(vi). The cone and CR branch remain inputs. This gives (viii) and the release boundary. □

Conclusion

v6 closes what the constant-profile record can close:

- the catalogue matches its executed source;
- the v4 Slavnov–Taylor failure was a convention error, and the derived row closes with an a-priori constant;
- the D01 obstruction is exact, and it is Mellin-null;
- a second apex moment is proved.

A closed row is not the complete Slavnov–Taylor identity. A constant profile is not the full counterterm basis. A Mellin value is not a prescription theorem. The cone and CR domain remain inputs.

O.1 on the cone = `NOT_CLOSED`, `bulk_symmetry_restored` = `false`.

A Self-adjoint apex dynamics and the readout analogy

A.1 Boundary form and flux

For a real radial Bessel expression

$$L_\nu = -\partial_r^2 - r^{-1}\partial_r + \nu^2 r^{-2}$$

in $L^2(r\,\mathrm{d}r)$, integration by parts gives the endpoint boundary pairing, up to the chosen outward orientation,

$$\mathfrak{b}(u,v) = \lim_{r\downarrow 0} r\left(\overline{u}\,v' - \overline{u'}\,v\right). \tag{A.1}$$

A self-adjoint domain makes this pairing vanish on every pair of domain vectors, with the appropriate sum if channels are coupled.

Proposition A.1 (No absorbing apex in the declared closed realization)**.** *Let A be a self-adjoint realization with the declared closed system boundary conditions. Then:*

(i) *$e^{-\mathrm{i}tA}$ is unitary and preserves Hilbert norm;*

(ii) *the integrated endpoint probability flux of sufficiently regular domain states vanishes;*

(iii) *a normalizable bound state is a state of that realization, not an absorbing output channel.*

Proof. The spectral theorem gives a unitary group $e^{-\mathrm{i}tA}$. Equivalently, for differentiable domain solutions,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u(t)\|^2 = 2\,\mathrm{Re}\langle u(t), -\mathrm{i}Au(t)\rangle = 0.$$

The integrated radial current is proportional to $\mathrm{Im}(r\overline{u}\,u')$. The vanishing of $\mathfrak{b}(u,u)$ gives zero net boundary flux. A bound eigenvector evolves by a phase $e^{-\mathrm{i}t\lambda}$; its norm is not lost. □

This is standard self-adjoint-extension theory [27, 26]. For a channel-coupled boundary law, the proposition concerns the total flux, not necessarily each component separately.

Remark A.2 (A negative bound state and wave stability). Self-adjointness alone does not imply that $\partial_t^2 u + Au = 0$ is a stable positive-energy wave theory. If $Au = -\sigma^2 u$, its wave component solves

$$\ddot{a} - \sigma^2 a = 0.$$

The sine quotient becomes $\sinh(\sigma t)/\sigma$. One must not call this positive-frequency evolution $e^{-\mathrm{i}t\sqrt{A}}$ without a nonnegative operator.

The F and stated scale-invariant CR Bessel realizations above are nonnegative in their declared range. A different non-Friedrichs extension with a negative bound state retains self-adjoint Schrödinger evolution but requires this separate wave-stability qualification.

Ghost and gauge-fixed tensor fields also do not automatically carry a positive probability interpretation on physical cohomology. No such identification is inferred from the radial boundary calculation.

A.2 What a loss channel would additionally require

For the same complete single-particle realization, an absorbing endpoint would require a different, dissipative boundary model rather than the self-adjoint one just analyzed. If the time evolution is written as e^{-tG}, its contraction generator is expressed by the appropriate maximal-accretive condition on G; the sign convention should not be transferred uncritically to a Hamiltonian written in another convention.

There is another possibility: couple the cone to additional quantum degrees of freedom and then restrict access to a subsystem. The complete enlarged dynamics may remain self-adjoint

and unitary, while its accessible reduced channel is lossy. Thus the unrestricted statement "every form of loss requires a non-self-adjoint Hamiltonian" would be false.

Neither an absorbing boundary condition nor an apex reservoir coupling is constructed by this paper. Both would change its declared model and require a new gauge/BRST analysis.

A.3 The precise I-C comparison

I-C distinguishes destructive finite readout from coherent return and from thermodynamic implementation [19]. Its six statements are relevant only as the following separations.

A: recovery metric.
If the complete accessible output of an unknown d-dimensional code is classical, with no additional message-bearing quantum resource available to the decoder, every deterministic decoder $\mathcal{D}$ obeys

$$F_e(\mathcal{DC}) \le \frac{1}{d}, \qquad \frac{1}{2}\|\mathcal{DC} - \mathrm{id}\|_\diamond \ge 1 - \frac{1}{d}.$$

The distinct Haar-average pure-state infidelity is at least $(d-1)/(d+1)$, using $F_{\mathrm{av}} = (dF_e + 1)/(d+1)$ [29, 30].

B: append-only entropy.
For one unconditioned retained history,

$$H(X_1, \ldots, X_{n+1}) = H(X_1, \ldots, X_n) + H(X_{n+1} \mid X_1, \ldots, X_n).$$

This monotonicity is not a Page curve for unmeasured quantum radiation.

C: conditional thermal accounting.
For an initially independent Gibbs bath and a complete unitary step,

$$\beta_{\mathrm{th}} Q_B = S_{\mathrm{n}}(\rho_M) - S_{\mathrm{n}}(\omega_M) + I_{\mathrm{n}}(M:B)_\omega + D_{\mathrm{n}}(\omega_B \| \tau_B).$$

A strictly positive complete non-bath entropy decrease implies positive heat. The equality itself does not require that decrease, and a readout channel does not fix the heat bill [31].

D: complete classical exterior.
Jointly classical heat labels and finite postprocessing do not evade the classical-cut recovery obstruction. Quantum access to the bath is a different interface.

E: endpoint.
A finite destructive prefix does not select physical evaporation, a remnant, or a last-moment change to a quantum output.

F: classical persistence.
Persistent observables commute when the retained accessible output is declared classical-only. Returned quantum residuals and prior transfers cannot be omitted from the complete account.

For completeness, the recovery obstruction follows from the measure-and-prepare form

$$\mathcal{DC}(\rho) = \sum_x \mathrm{Tr}(E_x \rho)\sigma_x.$$

Its referenced output is separable. For the maximally entangled state Φ_d, every product state has overlap at most $1/d$, since

$$\mathrm{Tr}[\Phi_d(a \otimes b)] = \frac{1}{d}\,\mathrm{Tr}(a^{\mathsf{T}} b) \le \frac{1}{d}.$$

Convexity and testing the Bell projector give the two displayed bounds. This argument concerns a finite accessible channel, not an apex extension.

A.4 Why the analogy supplies no O.1 premise

No entanglement-breaking map, classical-only radiation algebra, bath-reset process, or finite acquisition instrument is part of the cone residue calculation. Conversely, the self-adjoint apex realizations do not choose such an interface.

The correct conclusion is therefore:

I-C supplies no usable analytical input for cone O.1; self-adjoint endpoint data are not destructive readout; no Ward identity or apex pole follows from the analogy.

This is a separation, not a premise of any graph calculation and not a claim that reduced access loss is impossible in a different enlarged model.

B Source and status declarations

Analytical inputs. The Bessel heat representation, standard Bessel transform identities, self-adjoint functional calculus, and conventional dimensional regularization background are identified analytical inputs. The apex normalization, channel-shift integral, conditional reduction argument, and polynomial identities are proved explicitly.

Companion source scope. The supplied v4 source, supplied I-C sources, version-qualified excerpts, and programme descriptions are distinguished from complete independently inspected manuscripts. Companion manuscripts were not re-read for v6; their citations carry the mark †. No inaccessible source is claimed to have been read in full.

Primary literature. The bibliography identifies primary analytical and quantum-channel sources for attribution and normalization. The original shorthand citations "FM97" and "S11" are not used as substitutes for precise hypotheses. The normalized apex constant is proved here independently of those abbreviations.

Execution records (v6). Statements tagged [exec] were executed on the shared working machine and are logged. Software: Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0, and pdfTeX 3.141592653-2.6-1.40.26 (TeX Live 2025). The XVI-P/Q/R trees were hashed before and after this revision with the command of Appendix C; the digests are identical. No file in those trees was edited. Nothing was pushed or written to any site. The registry recheck read the public listing only. The v4-era statements that v6 did not rerun stay attributed to the supplied record.

Compilation and review. This source was compiled with two pdfLaTeX passes. It has an internal bibliography, no external figure, data, or bibliography file, and no unresolved references. No proof-assistant check, independent referee review, physical implementation, or journal acceptance is asserted. In particular, [exec] is not independent review.

Submission obligation. A submission making the full loop-coefficient claim must attach or archive the missing graph certificate and obtain the author's approval of the mathematical scope and source versions. The present conditional theorem must not be relabeled an independently reproduced complete computation; [exec] means reproducible on archived inputs, not independently reviewed.

Unchanged non-claims. The cone and CR extension are prescribed inputs. Companion connections and OpenAI-attributed catalogue records are documentary separations only. No catalogue proof is adopted or independently verified. O.1 on the cone stays `NOT_CLOSED`; `bulk_symmetry_restored` stays `false`. XVI-N is not discharged, the downstream XV-B admission exit is not opened, and the H2/I3/J2 analytical chain is not completed. No Einstein dynamics, physical tensor-core identification, spacetime emergence, or inherited characteristic quantum coefficient is constructed.

C Artifact manifest and rerun commands (new in v6)

Paths are relative to `/workspace/XVI-O1-CONE/`. `RO:` denotes the read-only copies under `/tmp/o1cone_ro/`; their digests equal those of the originals in the frozen trees. The frozen-tree digest is

```
cd /workspace/XVI-$t && find . -type f -not -path './.pytest_cache/*'
       -not -name '*.pyc' | sort | xargs sha256sum | sha256sum
```

Tree	SHA-256 (before and after v6)	before vs. after
XVI-P	ec965c15ae25cd3bb9cc2b3a134e966b 19a1803c78d1fe5bf7554da2a8b3aaa6	identical
XVI-Q	0df9a0d10409d31ac0c2d0a47197e3f6 e03c7ae258ec5fb775137f66cb0493c4	identical
XVI-R	57187bc6825f5c6ae88633b7279c6c0b e57f7c4918fdbc525da4aaf958331edb	identical

File	SHA-256
Base source (read-only copy)	
`XVI-O1-CONE-v5.tex`	e00e49f548bdc3dd1da94411e7438f73 c873aa3e2b3d24bae3d74f2bd3a6985b
Programs	
`code/spectral_apex.py`	cb22811e721c932766a576cb15626015 584f9cbc3cc3696f793e664c43debde2
`code/st_test_const.py`	dd87dfd55dbf1a730ad31d728fced6d9 aa1f35c59afc0b5ab0f8072f9b81f0ee
`code/st_scan.py`	4a03c966a77b60a028e57163aef7f264 5118c797ec3d55fc5c44ac8ee63cf759
`code/catalogue_check.py`	851f295b79cf2a483e6f9de4558d0f84 2082bd7eff46efd1866d2ad93c3180df
`code/st_v6.py`	5dba49edc69ff70ca01cb0a428aef450 262c117f19a6d86074acf4dc89cb6155
`code/st_norm.py`	da1c19a9f75926f26490d44129fc21fa 952b97384d4784a7c79219e5f1ff0f31
`code/st_v6_stats.py`	36ec924f70166b004720d547c1bf6cac 4ae2283aec8321ea7db2234639241c31
`code/flat_pole.py`	7d77f198d4cd6874c6a88f8bff554835 97b22ed74be4dc66bc7a2bbfd9423dc2
`code/flat_vs_p18.py`	4bee961081e64fa4ca7de67baa92bd51 5bbd3307acaf317390c9fa5c68912561
`code/weighted_apex.py`	a8a3ffd0742a5fdc5d176d8b259d4701 1051513ff48191fcc22231e94cb2eada
`code/weighted_checks.py`	a14d14e9112ab3603c4d77bbaed10cf1 dc70e67357ac4678d116de89eb317834

File	SHA-256
code/hi_d01.py	f3e7ee7f82bde0c6aa85d4fc57e2807a47fe95a53b9af5348d31b206a0e524d5
code/hi_d01_exact.py	c4e7d0271a0890d604e0ae1e8b300bb53bdaa64402e1188faf72247133d0f118
code/hh_structures.py	adc0bdbe0482e92d59bd26de05b90c41e5d90e7dcb390c6ef6440f9d3c3f3f23
code/build_v6.py	40cd45f3b622260020cfb6f52a91904fc2894a457be599fb12faec260b09663e
code/artifacts_v6.py	d602a6ab9b12105f02baa274c2c212c3624c1c30d96c490cc8bb95e5e9020d00
Spectral logs (v4 session, audited in v6)	
logs/v4/spec_D01_CR.json	ee0c4766a8282d4969366eb44070617f715b535fc72c3760eb47e80130419a3e
logs/v4/spec_D01_F.json	7ea9fe6077c7b28d1bbad1f25fa7325aa88e377227f87683d3a3ee570b9da819
logs/v4/spec_D01_UNIT.json	0923e4c51b68021f6fa77d734cdf25ecfe758385a542b26207f1c97819b686a6
logs/v4/spec_D02_CR.json	679ac21d06a9b8de0657f15fbcb116cd6fbfcf5af7a5e2e56abd99cf2bde6398
logs/v4/spec_D02_F.json	827cc09f0945b69a9d0cba1bacf523df9e0842474322072803418 2a364de5d65
logs/v4/spec_D02_UNIT.json	2f12785331b61aa9534515191e916f46cbba7d418c965b1e221140ab2bb36ee3
logs/v4/spec_D03_CR.json	6cae7e5ce69623b004ca177af9c4971b469976b704638810f978e68911a5e78f
logs/v4/spec_D03_F.json	7545563c038b73d232990a7b636055c14687293c67e3cad76ba9fdd703261ef9
logs/v4/spec_D03_UNIT.json	8aee1bbdc1f7e6c31a378ae418054220e605fd04314b9fc26e81b09ea34fa415
logs/v4/spec_D04_CR.json	af01b84ec1891fff4c95d0ff43fa085417ceaff7e876395026f3a506b84cbd72
logs/v4/spec_D04_F.json	7a209c8c669ca21a8119e168cfb40e62e0b9642939dd9ef27a7307b07e6f4a2b
logs/v4/spec_D05a_CR.json	c23d25b994b17fde9aa78a63bf52092733d452568a10ab7af77f828e4ddcbe03
logs/v4/spec_D05a_F.json	c56f0489caae6064ba22ee9de188a66e4b173b11648e1a051c62fb387a5f3d9e
logs/v4/spec_D06_CR.json	d759c4b910babacbc60b2aa2746da136669b2d40bbc3952e48c15bdd10f2dbcd
logs/v4/spec_D06_F.json	2fead6732f276e532fedc4d0aaf8bd2719b584ff7e5f23fa82701cef29c45652
logs/v4/spec_D06_UNIT.json	a75b8ab0791896a4d64c06d14724c7f1dccb5e0eed5358a6c40633b2190f360c
logs/v4/spec_D11_CR.json	6afa59f62437354f5dff58693c421e4b53867224bdd11b4b5ea47303f79be971
logs/v4/spec_D11_F.json	b30291938bc5878556b69e6a2f8f4b885ca803e5f3da1c0a60bb79727f1f55bb
logs/v4/spec_D11_UNIT.json	f099bb41ff2ccbcfeb0bd568f3daf002ec164a91d63ccf216b6aaf9347306508
logs/v4/spec_D12_CR.json	d5f4ef51fd1adcf1e163422b76dc34212a34a0c285cae006cfdda5fa1c2cc948
logs/v4/spec_D12_F.json	59937795d9674227954d36157ffb2d2808d38244e9346c55f416103b01c4abcc
logs/v4/spec_D12_UNIT.json	b92e6a75b387400af13d4357cd5039dc5ce58b98f3a7e0c531048d78f39be508
logs/v4/spec_D13_CR.json	322b2653b807c47465cf9972da6acf5ea0ef566b69374dfecb86ff402c497552

File	SHA-256
logs/v4/spec_D13_F.json	ef4ef5e3cef8f2e9339a82e4f9f9d823844a13c2ebb10256bed893c6fb4b8f15
logs/v4/spec_D13_UNIT.json	9670e118cdd948d839d714242ba142afe30cc1bf994c985d41ab2e8b583935a8
logs/v4/st_const_CR.json	23897227e71dea9022e56edf2441ac4256cbd97149036b3e46dccd3521ed64ff
logs/v4/st_const_F.json	56ca503cb7b1dd02db11bbc18658a373dabeaaa579e7b39baf3b8ebb9e4e680d
logs/v4/st_scan.json	e86875743f6e5c99852456e3d4b492930727f4e0724a4f7dd1111e2ae41628db
v6 logs	
logs/v6/catalogue_check.json	1a436314aaf0c5a7ca7f8cc7a7a09be0a30394d80a1bbe2023fae6cca35f3e53
logs/v6/flat_D01.json	29293d0cf86d9a9091df2b6c5c613ea9df2b501563164eca1ad4bd0db62dc86d
logs/v6/flat_D02.json	80804ff73c1b02d05f0b720df14b957dfb73a38e0a9adeaa717e532bc1fcdf84
logs/v6/flat_D03.json	ff09567fc9a68e73aa0b5c963359b1118445d807a9707e37c8b087b4dd0c31a9
logs/v6/flat_D06.json	346f4bf0cb4072a5f8cd38e2c725c06963bce2f8714ca50d7d92b7bae81d7332
logs/v6/flat_D11.json	9137e362feca3054545742d7cea4e4399fecddd117bdce0093bb79cb3705dbec
logs/v6/flat_D12.json	80804ff73c1b02d05f0b720df14b957dfb73a38e0a9adeaa717e532bc1fcdf84
logs/v6/flat_D13.json	a8008a4a8ac47c9fac9c39efa5729dd74fa9d7e70aae824e78407def473578d1
logs/v6/flat_vs_p18.json	891bf4a356e9427dea75d403874ec4220d4fa5ba74de59b86ee8152bb010c42c
logs/v6/hh_structures.json	78fb8461a611975d88dc600ea2f6e937b52a31e62b6ed0483fbf721458b5086a
logs/v6/hi_d01.json	1ee5a24338961e2fb35006fb00ebee020ad93ab20bd78c8f5c0dbe2b7d3dbbb1
logs/v6/hi_d01_exact.json	850e69e35f7d33cf0933886869cc65f5cd1fdfff1970d8fe1e2bba8ea7979305
logs/v6/st_norm.json	5b30e5af54e2a46f48f45356d2dee5434bc378fd19c3a1b4ee5e98fb5f197253
logs/v6/st_v6_CR.json	9f1065ca1c9ae07f4eb640061540e50ecad8faa0ebc75afd097d5fd943cf8dda
logs/v6/st_v6_F.json	eb09490b5b839a503d9a053fd5d5180a3473c4507896ef01dd79a7322b23be22
logs/v6/st_v6_FLAT.json	46b533ea296208474c3893d989fd6c45b3a874acd4e15a92b49e29fc4896bacb
logs/v6/st_v6_UNIT.json	20a9d7d327e92d8f182e54954aa7bd1d29cc8995bc35a9803ac239401001be07
logs/v6/st_v6_stats.json	bde0de688cc0e7317f3d42a2cf04975859d747c16e00c0cfe66cc4bda2ca7711
logs/v6/weighted_apex.json	b23bf74962e3174c689db732c5a4603c2d380e2be75c58a324dd6a337ead0c2d
logs/v6/weighted_checks.json	bc1b45f39bc41eadb640478f406b265551d63dea14a53017d5350d8a582fced5
Read-only frozen inputs	
RO:XVI-P/code/p5_entry/amplitude_rules.py	0daf63a3c8803d54514cb2aac29bea127c379e025f32c5b0219d365de5bdde4a
RO:XVI-P/code/p5_entry/feynman.py	4fab8b0d4a1cc0b4cd0be19ec30ace80559b92fa9808f1fc25c858599a389344

File	SHA-256
RO:XVI-P/code/p5_entry/dred.py	24112d9a6584613474e96f538840a9ff 6aecbf9ebd588cd890fd38d2e295de5a
RO: XVI-Q/provenance/XVI-P/certificates/P18_D11.json	af696a2ad7ec85fbba966e5403079c65 f5635d704ec4814f7755ff256d2031e1
xviq_regen/Q02_bulk_graphs.json	4ac6dc39f6e1763f1a90bceb45296fab d539e086c160d3ea60427350cf118505

Rerun commands. All commands run from `/workspace/XVI-O1-CONE`. The spectral runs were launched as detached jobs with logs because the machine has about 7 GB of memory.

```
for g in D01 D02 D03 D04 D05a D06 D11 D12 D13; do for b in CR F; do
  setsid python3 code/spectral_apex.py $g $b > logs/v4/spec_${g}_${b}.log 2>&1; done; done
for g in D01 D02 D03 D06 D11 D12 D13; do setsid python3 code/spectral_apex.py $g UNIT \
  > logs/v4/spec_${g}_UNIT.log 2>&1; done
python3 code/catalogue_check.py          # -> logs/v6/catalogue_check.json
python3 code/st_norm.py                  # -> logs/v6/st_norm.json
for b in CR F UNIT; do python3 code/st_v6.py "logs/v4/spec_{g}_$b.json" $b; done
for g in D01 D02 D03 D06 D11 D12 D13; do python3 code/flat_pole.py $g; done
python3 code/st_v6.py "logs/v6/flat_{g}.json" FLAT
python3 code/st_v6_stats.py; python3 code/flat_vs_p18.py
python3 code/weighted_apex.py; python3 code/weighted_checks.py
python3 code/hi_d01.py; python3 code/hi_d01_exact.py
python3 code/hh_structures.py
python3 code/build_v6.py && python3 code/artifacts_v6.py
cd v6build && pdflatex XVI-O1-CONE-v6.tex && pdflatex XVI-O1-CONE-v6.tex
```

The spectral runs (first four lines) were executed during the v4 session and were not repeated for v6; every other command was run for v6.
Marks: * read in full for this revision (for programme records: the specific files named); ‡ bibliographic details confirmed online; † cited from memory or from earlier supplied records, not re-read here.

References

[1] * N. Cheung, *Scoped O.1 on the Cone, v4: Spectral Apex Reduction, Exact Constant-Profile Apex Poles of All Two-Point Graphs, the Ghost-Equation Row, and an Unvalidated hh Slavnov–Taylor Test*, 18 October 2026. Supplied complete source `XVI-O1-CONE-v4.tex`. Read in full. Its code and logs are now archived and audited (Appendix C).

[2] * N. Cheung, *Scoped O.1 on the Cone, v5: Spectral Apex Reduction, Constant-Profile Pole Records, and Ward-Identity Boundaries*, 18 October 2026. Source `XVI-O1-CONE-v5.tex` (SHA-256 `e00e49f5...a6985b`, full digest in Appendix C), read in full; the base of this revision.

[3] * N. Cheung, *XVI-P*, frozen p22 contract as described in the supplied cone source; separately supplied p23 excerpt retaining the p21 absolute-D11 normalization and scope discussion. Enter Calculus Companion, unpublished manuscript records. The later excerpt is a comparison, not a silent contract update. Files read in full for v6 (read-only copies, digests equal to the originals): `code/p5_entry/amplitude_rules.py`, `feynman.py`, `dred.py`. Relevant scope locators include `rem:o1`, `rem:p5`, and `rem:qds`.

[4] * N. Cheung, *XVI-Q*, supplied Q02 normalization description and separately scoped Q11 release record. Enter Calculus Companion, unpublished source records. v6 reads the regenerated Q02 graph record `xviq_regen/Q02_bulk_graphs.json` in full and the read-only

certificate `provenance/XVI-P/certificates/P18_D11.json`; the Q11 scope is retained as supplied.

[5] † N. Cheung, *XVI-R*, frozen read-only comparison record identified in `XVI-O1-CONE-v4.tex`. Not read for v6; only its full tree digest is recorded (unchanged before/after).

[6] † N. Cheung, *I-B: A Finite Enter Spacetime: Calibrated Destructive Time Readout on a Prescribed Conical Product*, version 1. Supplied version-qualified excerpts, Enter Calculus Companion. Relevant comparisons: `def:experiment`, `thm:instrument`, `prop:interfaces`, and `prop:not-source`. https://enter.grok.me/browse/I-B.

[7] † N. Cheung, *En09-A: Realization of Relational Time by Destructive Enter on a Clock Register*, version 1. Supplied excerpts, Enter Calculus Companion. Complete finite record–residual comparison only. https://enter.grok.me/browse/En09-A.

[8] † N. Cheung, *I-B-G: Intrinsic and holonomy-twisted Green operators on a prescribed conical spacetime: Self-adjoint extensions, diffraction, finite propagation, and supported transfer from flat-completion laboratories*, version 2. Supplied source descriptions and Green-domain excerpts. Comparisons include `thm:green`, `eq:Carslaw`, `thm:transfer`, and `rem:compact-exact`. https://enter.grok.me/browse/I-B-G.

[9] † N. Cheung, *I-B-G-S: Intrinsic spin-1 and spin-2 Green operators on the prescribed conical spacetime*, version 2. Supplied unpublished excerpts and version-qualified descriptions, including `thm:spin1`, `thm:spin2`, `def:geo`, and `thm:selection`. Free source-test compatibility is not interacting loop closure.

[10] † N. Cheung, *I-B-G-S-B*. Supplied programme description of positive tensor-core selection, exceptional source-test defects, matching, and analytical kernel enclosures. Comparison locators: `prop:warddefect`, `cor:exclude`, `thm:contrast`, `lem:modetail`, `lem:complexbound`, and `thm:Cauchy`. A complete version-qualified proof source was not supplied for independent audit.

[11] † N. Cheung, *I-B-ECHO-v2: Diffracted echoes and finite supplied-report instruments*, version 2, October 2026. Supplied full-source record and source-qualified comparisons. Relevant locators: `lem:front`, `thm:gtd`, `thm:readout`, `thm:TV`, `thm:core`, and `cor:nonreconstruction`. No continuum acquisition or loop-residue theorem is imported.

[12] † N. Cheung, *I-B-VP: Vacuum polarization of the Friedrichs ground state on a prescribed conical spacetime*, version 3 and supplied version-qualified predecessor passages. Enter Calculus Companion. Comparisons include `lem:infrared` and `thm:friedrichs`. One-point state control is not the present loop UV/IR analysis.

[13] † N. Cheung, *I-B-VP-1: Prescribed apex finite parts on a conical Enter spacetime: Wick square, stress tensor, and charged current: strong extension, Ward identities, and classified contact freedom*, version 3, October 2026. Supplied excerpts and source descriptions. One-point distributional extension freedom is not a replacement for a spin-2 operator-domain selection.

[14] † N. Cheung, *I-B-VP-5-Ex1-B*, version 1, 18 October 2026. Supplied normalization, finite-part non-universality, and verification-scope excerpts. Relevant locators include `thm:running`, `thm:continuity`, `thm:equalradius`, and `thm:trace`. The historical v4 intake is not independently re-executed here.

[15] † N. Cheung, *I-B-VP-5-Ex2: Smooth flux-core selection on a prescribed cone: Scalar Friedrichs universality, Dirac-window selection, and quantitative exterior matching*, version

1, 18 October 2026. Supplied source excerpts and bibliography. Operator selection and Dirac-square comparisons only.

[16] † N. Cheung, *I-B-VP-5-Ex2-B*, version 1, 18 October 2026. Supplied excerpts concerning scalar Mosco selection, exhausted-annulus arguments, endpoint continuity, and finite-core qualifications. No spin-2 CR selection or loop Ward theorem is imported.

[17] † N. Cheung, *X-J-ST*, version 1. Supplied unpublished signal and clock instrument excerpts, including `thm:signal`, `thm:clock`, and `thm:TV`. Supplied-value reporting and complete failures are retained.

[18] † N. Cheung, *X-K-ST*, version 1. Supplied unpublished clock-composition and resource-fit excerpts, including `def:grade` and `thm:release`. A finite typed tree is not automatically a realized laboratory or a continuum acquisition process.

[19] † N. Cheung, *I-C-v2: The Shredder Interface: Destructive Readout, Black-Hole Analogues, and Quantum Computers*, October 2026. Supplied source in the revision material. Theorems A–F distinguish referenced recovery, append-only entropy, conditional thermal balance, classical exterior access, endpoint underdetermination, and accessible classical persistence. Documentary separation, not an O.1 premise.

[20] † N. Cheung, *XIV-H*, version 1; *XIV-I*, version 1; and *XIV-J*, version 1. Supplied excerpts. Historical H2 ending, constructed I2 internal amplitude, unresolved I3/J2 full characteristic map, and no inherited quantum admission. A partial off-characteristic reference is not the complete map.

[21] † N. Cheung, *XV-B*, version 2. Supplied technical-interface excerpts: reconstruction descent, relative lifting, normal-jet null tests, and conditional extension. The technical manuscript is distinguished from the unopened downstream quantum-admission exit.

[22] ‡ D. V. Fursaev, "The heat kernel expansion on a cone and quantum fields near cosmic strings," *Classical and Quantum Gravity* **11** (1994), 1431–1443. arXiv:`hep-th/9309050`. Apex-local heat coefficients and product-cone context.

[23] ‡ D. V. Fursaev and G. Miele, "Cones, spins and heat kernels," *Nuclear Physics B* **484** (1997), 697–723. DOI: `10.1016/S0550-3213(96)00631-1`. arXiv:`hep-th/9605153`. Spin-dependent conical heat coefficients and nonuniform smooth limits; not a substitute for the present graph-specific domain contract.

[24] ‡ D. V. Vassilevich, "Heat kernel expansion: user's manual," *Physics Reports* **388** (2003), 279–360. DOI: `10.1016/j.physrep.2003.09.002`. arXiv:`hep-th/0306138`. Heat-kernel normalization and Laplace-type coefficient background.

[25] ‡ F. W. J. Olver et al., editors, *NIST Digital Library of Mathematical Functions*, Chapter 10, especially Sections 10.29, 10.32, and 10.43 (10.43.22 confirmed online for v6). Bessel recurrences, modified-Bessel relations, and integral transforms. https://dlmf.nist.gov/10.

[26] ‡ J. Dereziński and S. Richard, "On Schrödinger operators with inverse square potentials on the half-line," *Annales Henri Poincaré* **18** (2017), 869–928. DOI: `10.1007/s00023-016-0520-7`. arXiv:`1604.03340`. Homogeneous Bessel realizations, spectral structure, and self-adjoint endpoint domains.

[27] † B. S. Kay and U. M. Studer, "Boundary conditions for quantum mechanics on cones and fields around cosmic strings," *Communications in Mathematical Physics* **139** (1991), 103–139. DOI: `10.1007/BF02102731`.

[28] ‡ D. Stöckinger, "Regularization by dimensional reduction: Consistency, quantum action principle, and supersymmetry," *Journal of High Energy Physics* **2005**(03) (2005), 076. DOI: 10.1088/1126-6708/2005/03/076. arXiv:hep-ph/0503129. Regulator consistency is distinguished from automatic preservation of a particular identity.

[29] ‡ M. Horodecki, P. W. Shor, and M. B. Ruskai, "Entanglement breaking channels," *Reviews in Mathematical Physics* **15** (2003), 629–641. DOI: 10.1142/S0129055X03001709. Preprint title: "General Entanglement Breaking Channels," arXiv:quant-ph/0302031.

[30] ‡ M. A. Nielsen, "A simple formula for the average gate fidelity of a quantum dynamical operation," *Physics Letters A* **303** (2002), 249–252. DOI: 10.1016/S0375-9601(02)01272-0. arXiv:quant-ph/0205035. The average-fidelity/entanglement-fidelity relation is attributed there to earlier Horodecki work.

[31] ‡ D. Reeb and M. M. Wolf, "An improved Landauer principle with finite-size corrections," *New Journal of Physics* **16** (2014), 103011. DOI: 10.1088/1367-2630/16/10/103011. arXiv:1306.4352. Conditional finite-reservoir entropy balance, not a heat cost deduced from the abstract readout channel.

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